GE VERNOVA

Operations

2.4.1

User Guide

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Change History

Date	Change Description
June 20, 2025	Operations 2.3: • Updated DER Attributes • Updated Measurement and Verification
June 05, 2024	Operations 2.1: • Added User Interface Overview • Added Automation Configuration • Updated Group Management

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1. Solution Overview

1.1. Introduction

The *Operations* module provides you with operational DER information to aid making data-driven decisions about how to improve your processes and how to leverage available DER flexibility to maintain the balance on the grid. Operations is a single-pane view that provides situational awareness, alerts, and the ability to manage grid operational issues.

With *Operations* you can aggregate the services of individual DERs into a more manageable, smaller number of virtual resources. This simplifies grid-related services because you can avoid the granularity of individual DERs.

The module enables optimization of DER utilization and existing infrastructure in various physical locations to get a cost-optimized and power-quality optimized outcome.

Operations functionality enables you to use flexibility to manage grid operational issues.

1.2. Logical Network Model Tree Viewer

A Logical network model (LNM), is a broad hierarchical description of the distribution system, which contains little of the connectivity model but allows you to understand the relative location and connection of a feeder or what substation the feeder draws power from. The Operations module is equipped with a tree viewer that allows you to navigate levels of the grid topology to view the relative position and connection of various grid devices and DER assets.

Operations is natively integrated with Gateway and pulls information from Gateway to populate the tree viewer, thereby allowing operation with a single source of input data.

You can upload the logical network model in Gateway. Operations will automatically synchronize the logical network model and be continuously in sync with any changes.

Once the model is available in Operations, you view it in a hierarchical topology tree format, where you can expand and collapse parts of the tree to home in on the individual assets on each feeder in the network. The top level represents the whole network. Expand the network to decide the next level to view, for example, a region, location, an individual feeder, and finally the device or asset itself.

Assets that can be included in an analysis have a checkbox next to the asset name. If there is no checkbox, the asset is unavailable for inclusion in an analysis.

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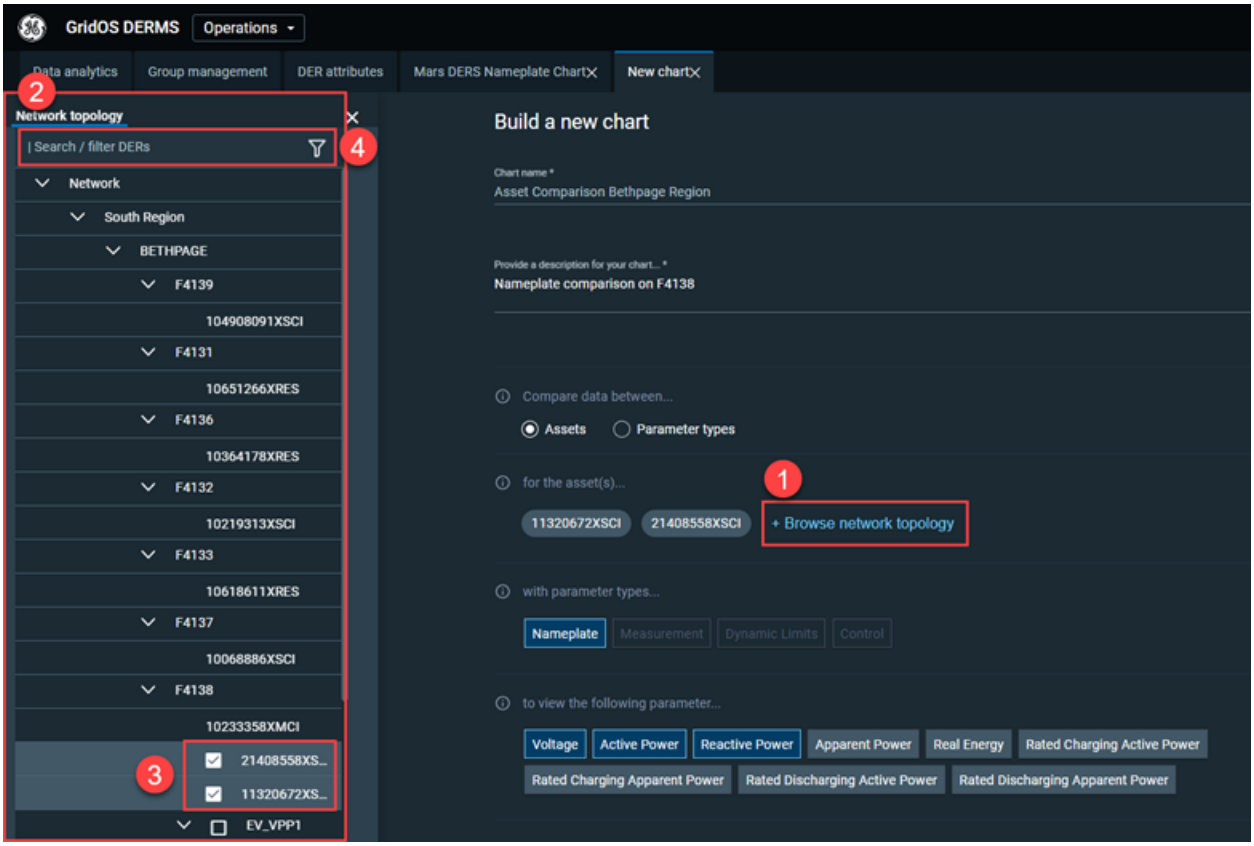


Figure 1. Build a new chart

1. Click on the **Browse network topology** link opens the topology tree in a left side pane.
2. Network topology tree.
3. Assets available for selection have a checkbox. Selected assets appear in the chart builder.
4. Assets on the network can be filtered by name by clicking the Filter icon .

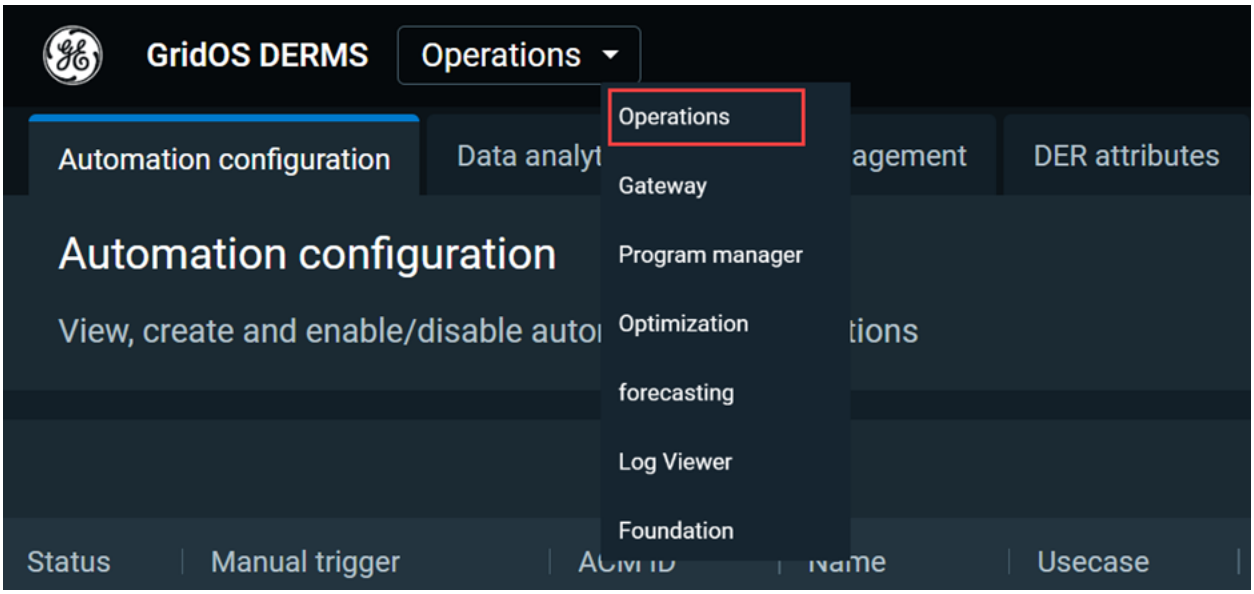
1.3. Geo View

Geo View provides DERMS Operators with a map-based interface to visualize DERs, and the corresponding operational states; support situational awareness and decision-making. A map-based workspace to visualize your network and assets, navigate by system hierarchy, and quickly find what you need. Designed for Operators, Planners, and Analysts who need situational awareness of the electric network and connected resources.

2. User Interface Overview

2.1. Operations Module

Once you are logged in, you can select a module from the modules menu at the top of the page. Only modules you have access to will be available from this menu.



The Operations user interface consists of the following tabbed pages:

- **Automation Configuration**: In this tab, you can view, create and enable/disable automation configurations, all the created configurations are displayed in a data table.
- **Data Analytics**: In this tab, you can build charts that will compare asset data on the network, or compare DER parameters, such as measurements, controls, or the nameplate data.
- **Group Management**: This tab allows you to group DERs and apply rules to the group to make it simpler to manage.
- **DER Attributes**: This tab is a tabular view of DER attributes and their details. You can select a group to view the details for each DER in the group.

2.2. Role-Based Access to Operations User Interface

The Operations UI is configured to ensure that only authorized users can access the user interface and perform sensitive operations, such as modeling and dispatch. The role-based UI access rights are listed in the table below:

Roles	Permissions					

	Grouping Read (View groups and DER attributes)	Grouping Write (View groups and DER attributes)	Automation Read (View groups and DER attributes)	Automation Write (View groups and DER attributes)	Trend Read (View groups and DER attributes)	Trend Write (View groups and DER attributes)
Analyst	Yes	Yes	Yes	N/A	Yes	Yes
Supervisor	Yes	Yes	Yes	Yes	Yes	Yes
Read-Only	Yes	N/A	Yes	N/A	Yes	N/A

2.3. General Operations

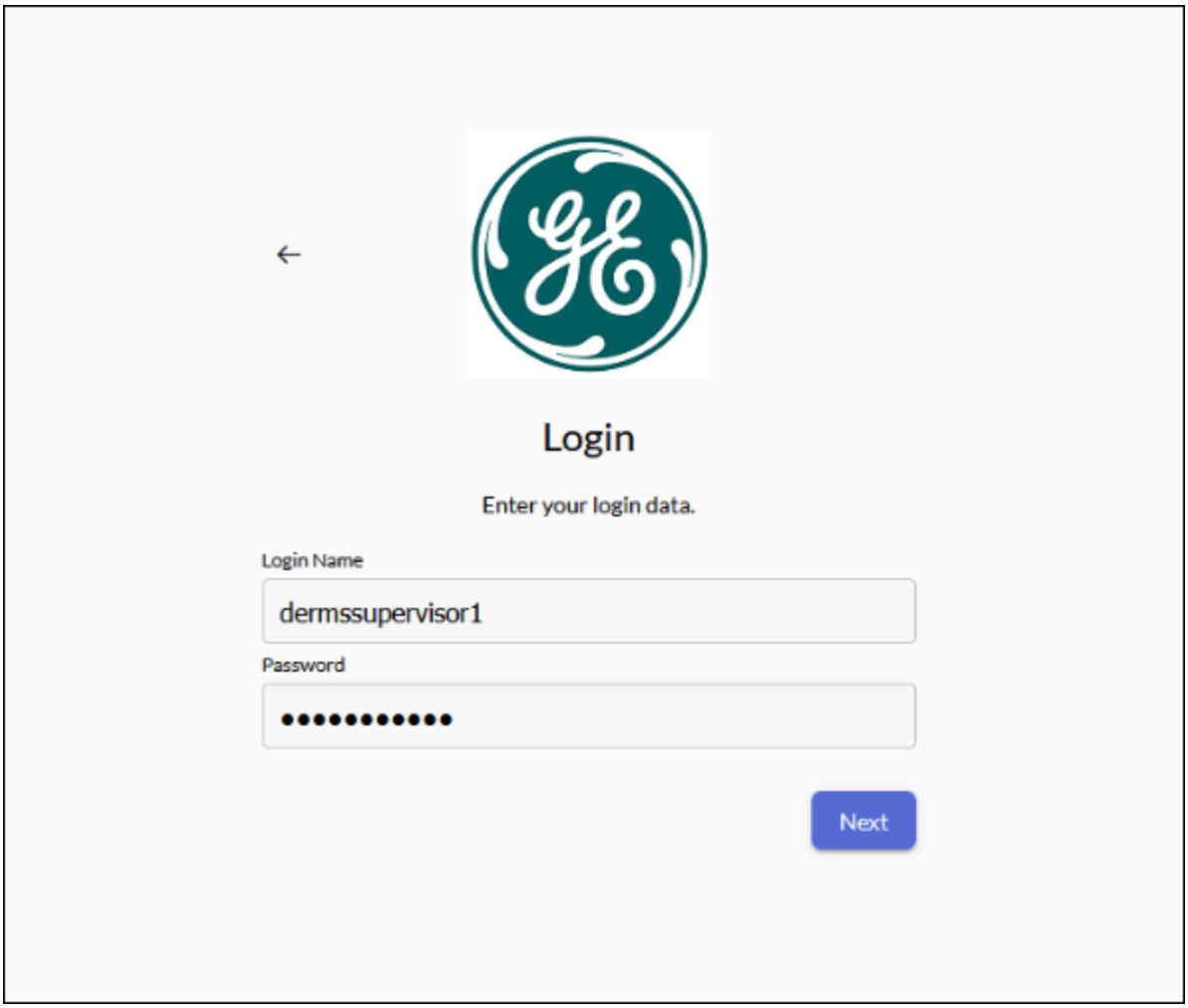
This chapter describes how to perform the most common activities in the Operations user interface.

2.3.1. Logging In

The module UI can be accessed using Google Chrome.

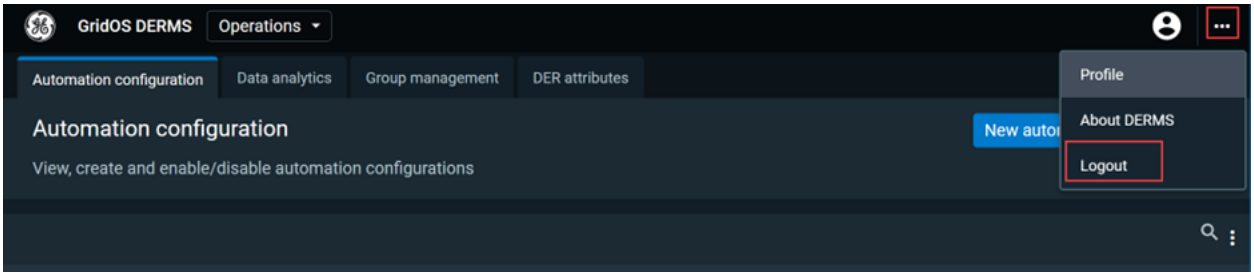
Your administrator will provide the URL and login credentials to your instance of GridOS DERMS.

1. Open your Web browser,
2. Navigate to GridOS DERMS UI. The login page opens.
3. Enter your username and domain, and then click **Next**.
4. Enter your password, and then click **Login**.



2.3.2. Logging Out

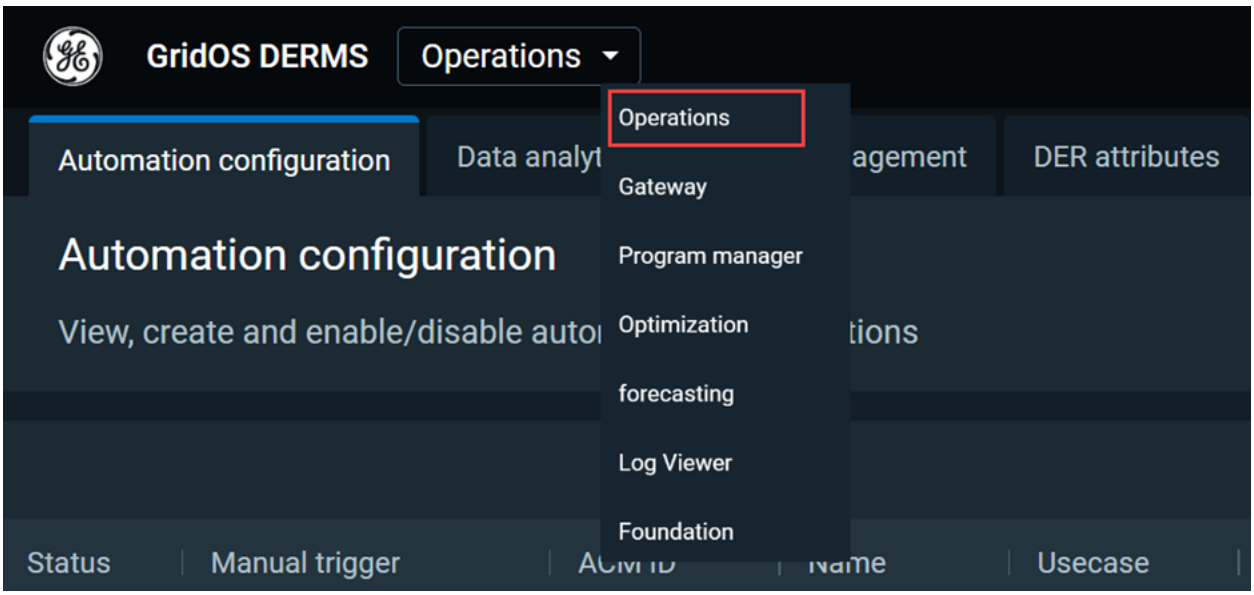
To log out of DERMS, click the ellipsis at the top right of the screen and select **Logout** from the menu options.



2.3.3. Launching Operations User Interface

Once you are logged in, you may select a module from the modules menu at the top of the

page.Only modules you have access to will be available from this menu.

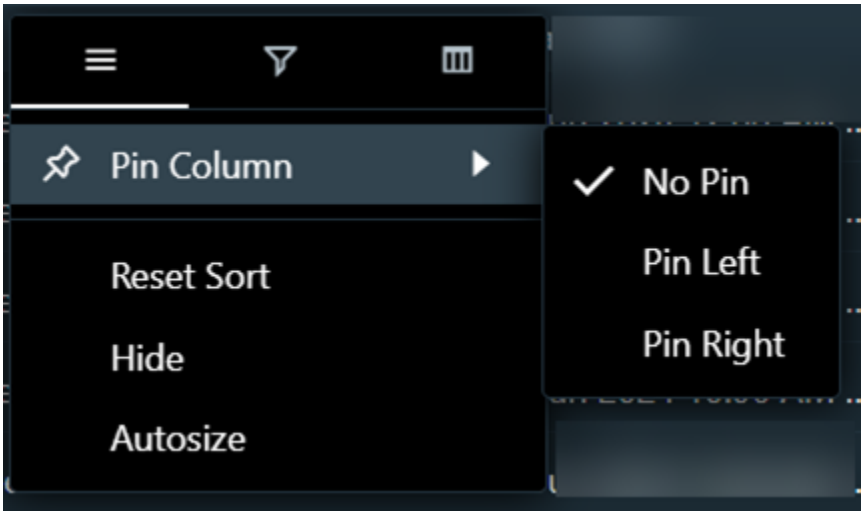


2.3.4. Customizing the Data Table View

Pin, Reset, Hide and Autosize a Column

To pin a column in order to set its relative position in the table:

1. Point at a column header, click the **Filter** icon at the right of the column header, and click the Options tab .
2. Select **Pin** and then the required pin option.
3. Select **Autosize** to automatically resize the width of the current column to fit the column content.
4. Select **Reset Sort** to reset the data and sort the column.

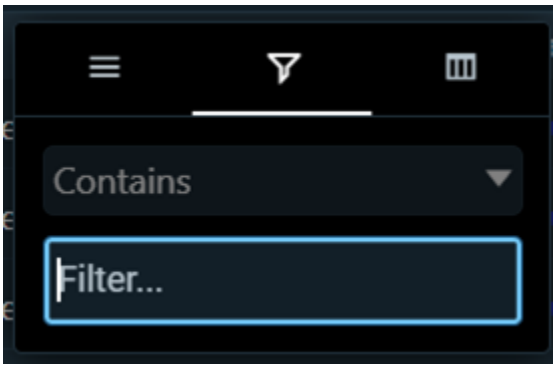


Filtering Data

You can filter the data in the to find specific records. Filter icon appears when you hover over a column heading.

To show the per-column filtering pop-up:

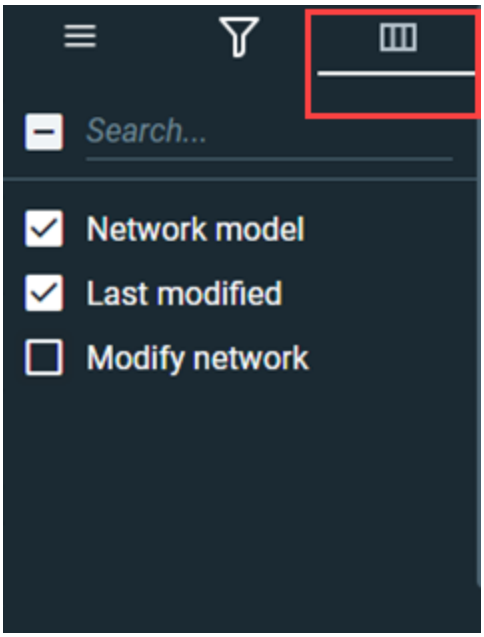
1. Point at a column header, click the **Filter** icon at the right of the column header, and click the Filter tab from the options shortcut menu.
2. From the shortcut menu, select a filtering criterion.
3. In the text box, enter a keyword to search. The table is filtered and displays records containing the keyword. Combine filtering criteria to narrow the search results.



Column Visibility

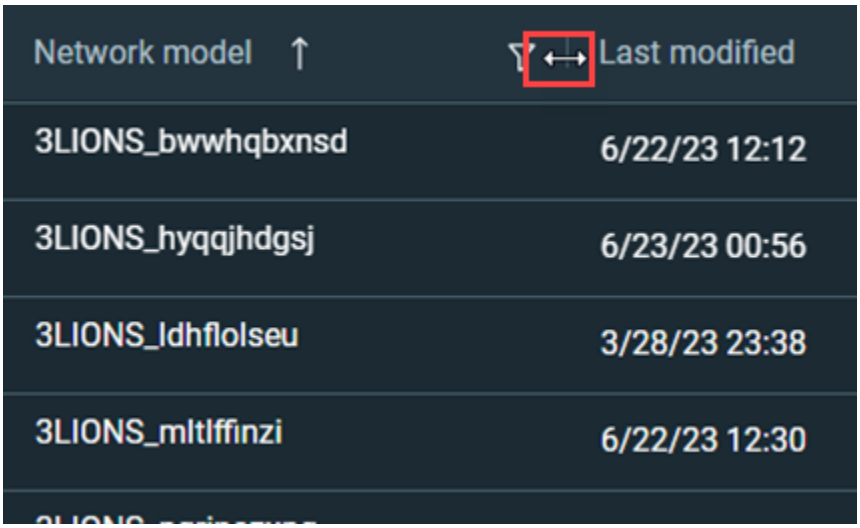
To define column visibility

1. Point to a column header, click the **Filter** icon at the right of the column header, and click the **Column** tab from the shortcut menu.
2. Select or clear columns, as desired.



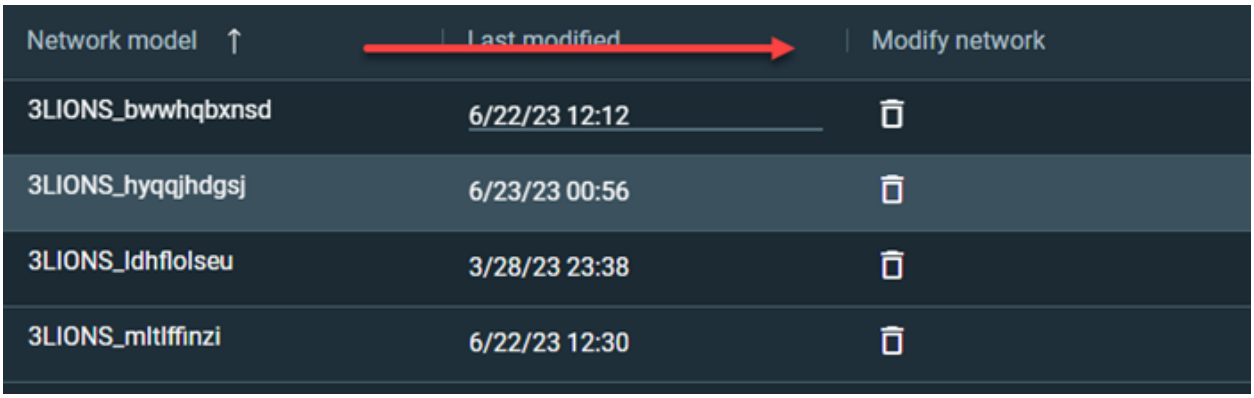
To set the column width:

1. Point the mouse to the column header border. A double-sided arrow appears.
2. Drag the arrow to adjust the column width.



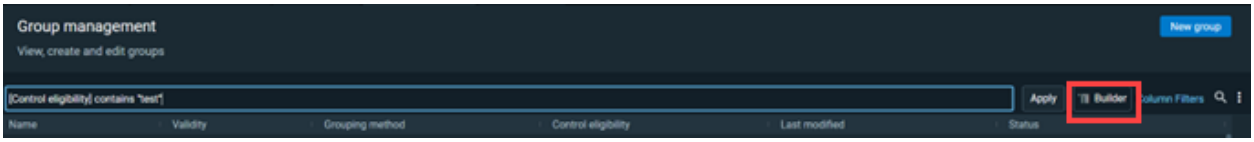
To change the column order:

Drag the column header to a different position.



Query Builder

Select **Query Builder** icon to create queries and filters.



3. Interoperability

The *Operations* module uses APIs to retrieve data from external systems. You can get the API specifications for *Operations* in this location:

`https://<aws-url>/operations/api/v1/openapi.json`

1. Log in to Operations.
2. Append `api/v1/openapi.json` to the URL and press ENTER.
3. Copy and paste the code into an API platform, such as Postman, for example, and configure appropriately.

   NOTE

   <aws-url> refers to your environment's domain.

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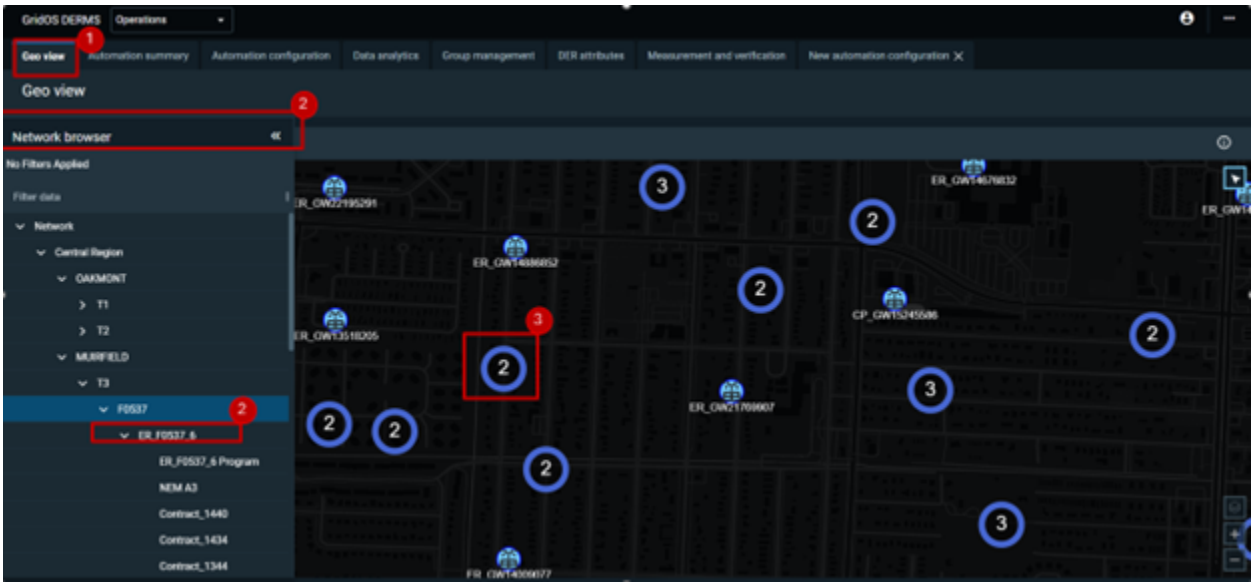
4. Geo View

A Geo View for Distributed Energy Resources (DERs) is an interactive, map-based interface that gives Operators and Planners situational awareness of the network and connected DER assets—such as Solar PV, Wind, Battery Storage, EV chargers, and controllable loads—by overlaying asset locations, status, telemetry, and events on a geographic basemap. Users can navigate via a side panel aligned to the network hierarchy (for example, region, substation, feeder), clicking a level to center the map on the relevant area and assets. To populate Geo View in Operations, upload the network model in Model Manager or Gateway and make sure that registered DERs have valid Latitude and Longitude values populated in Model Manager.

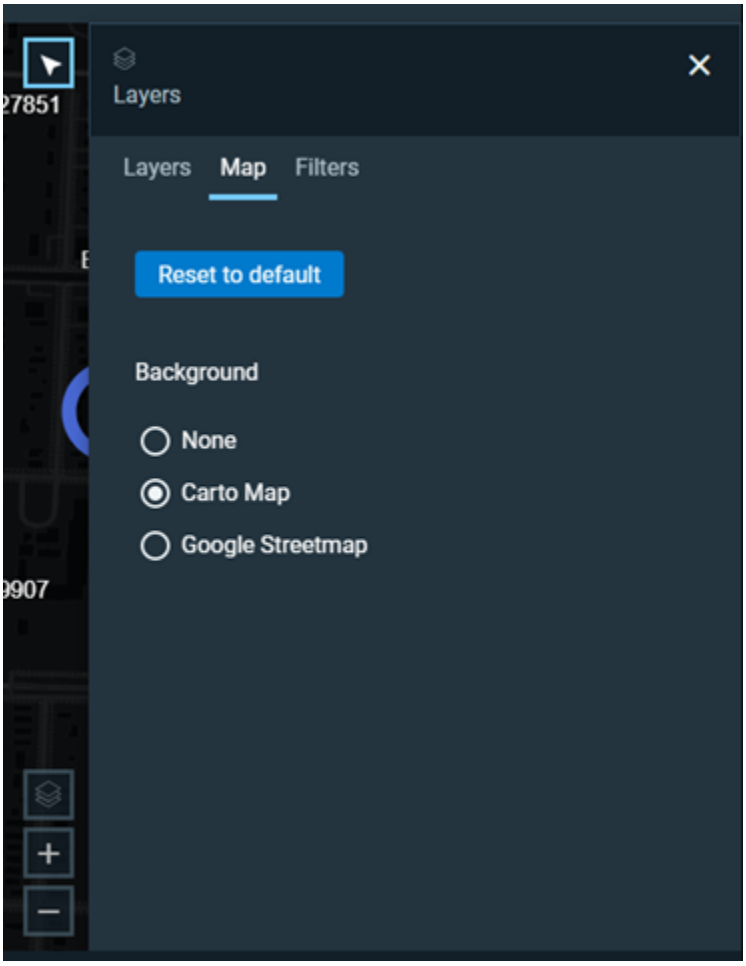
4.1. Navigating Geo View



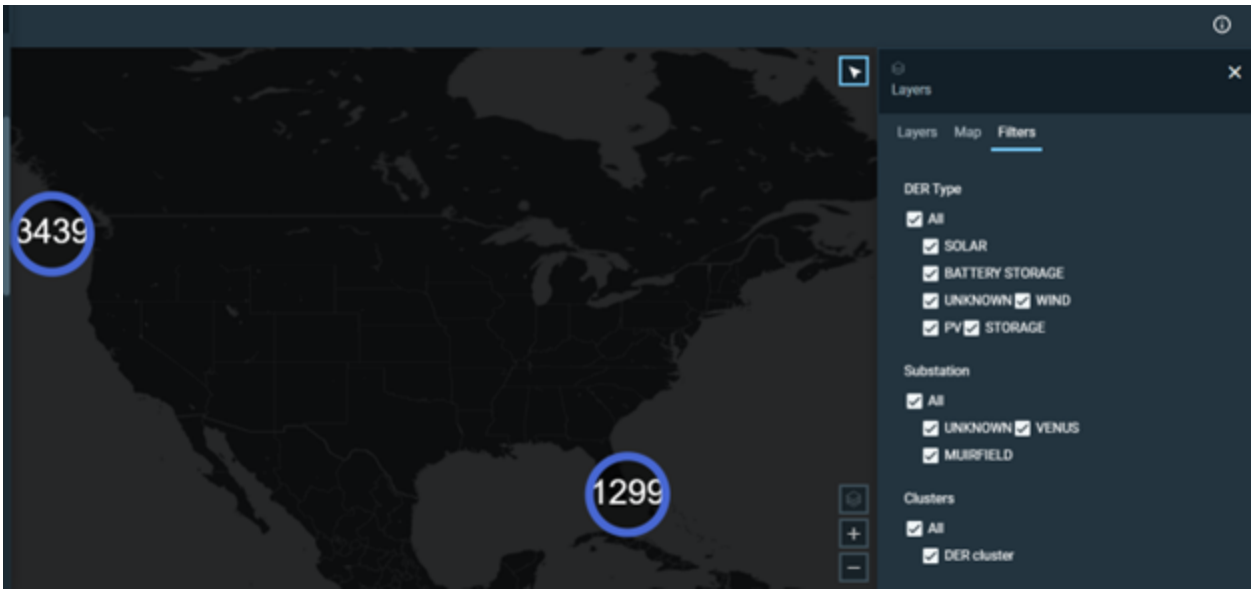
1. Open Geo View. The map will load with default layers and extent.
2. Use the side panel to jump to a part of the network and center it, or pan/zoom the map directly. Use the side navigation panel filter function to quickly filter DERs Pan: Click-drag on the map.
3. When you see clustered icons, zoom in to expand and see individual assets Zoom: Mouse wheel to zoom; double-click to zoom in. Use +/- buttons if available. Following list of DER types and corresponding Icons are supported currently by the Geo View

DER type	Icon
Battery energy storage	
Combined heat and power	
Combined PV and storage	
Concentrated solar	
Electric vehicle	
Electric vehicle supply equipment	
Flexible load	
Fuel cell	
Generic	
Hydro	
Micro turbine	
Other	
Solar PV	
Standby generator	
Virtual or aggregated DER	
Wind	

4.2. Geo View Map and Filters



Switch between different map backgrounds such as Carto Map or Google Streetmap in the Map selection in the layers side panel.

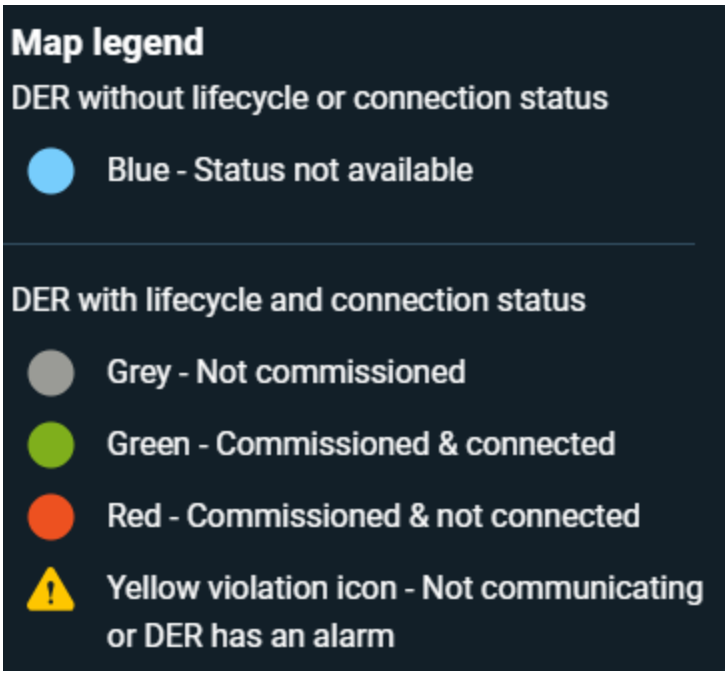


To filter the Map view use the filter tab. Following filter criteria are available based on map zoom level

1. DER type – Based on the DER types available in the “visible map”
2. Substation – Based on substation Name

4.3. DER live Telemetry and Status updates

Dynamic updates are published to Geo View for DER lifecycle status, telemetry as well as DER status. The map uses a combination of color and icons to represent different DER lifecycle status and communication status for a DER as per the legend below.



Clicking a DER icon on the map opens a DER info side panel (available release 2.4.1).

The side panel has the following sections and information:

1. DER metadata – meta data associated with the specific DER
2. Measured values – Live status and telemetry data received from the DER

3. Rated Values – Name plate rating data as communicated / registered for the DERs
4. Users can also from the “side panel” directly navigate to gateway manual control screen by clicking the manual control button in the side panel
5. The Active settings tab shows the currently active settings similar to the DER attribute Tab

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5. Automation Configuration

Automation Configuration identifies and resolve real-time forecasted constraints using DER. It integrates Forecasting, Operations, Optimization and Gateway into an integrated solution to allow utilities to incrementally test and roll-out closed loop control across the network.

You can view, create and enable/disable automation configurations. In this tab, you can view all the configurations created in a data table.

5.1. About Automation Configuration tab

The optimization analysis on a problem formulation is executed by configurable triggers. Active triggers can be run on any problem formulation any number of times.

This tab lists the status, manual trigger option, name, trigger condition. You can manually run a trigger by clicking the **Run** button from the table by using the slider next to a trigger to toggle it on or off.

> **Note**
>
> The **Run** button will be enabled only for active triggers. Disabled trigger will have the **Run** button also disabled. On clicking, the **Run** button appears in disabled status until the job request is finished and the notification appears on screen.

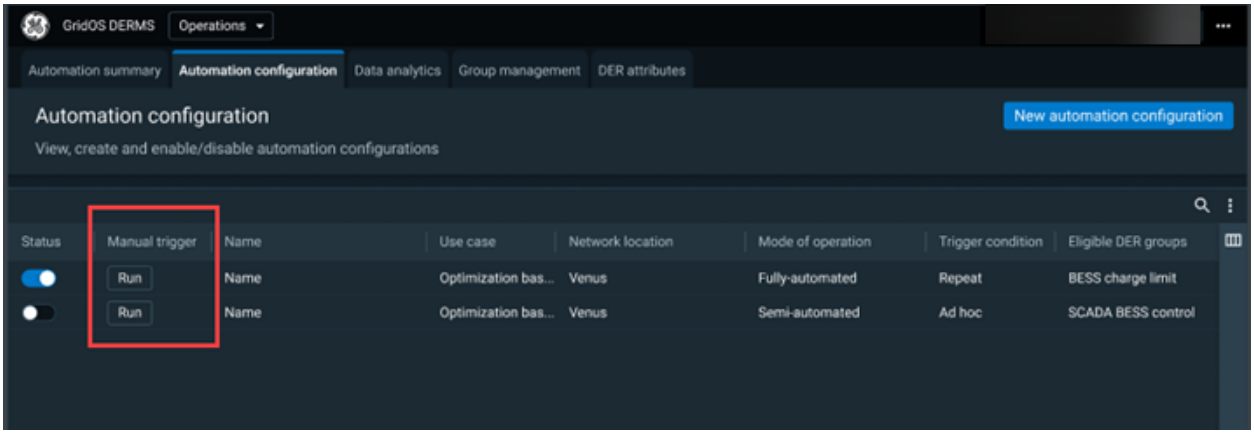


Figure 2. Manual trigger column

To define column visibility:

Click on the table-menu icon to sort, autosize, print and export the data via csv or excel.

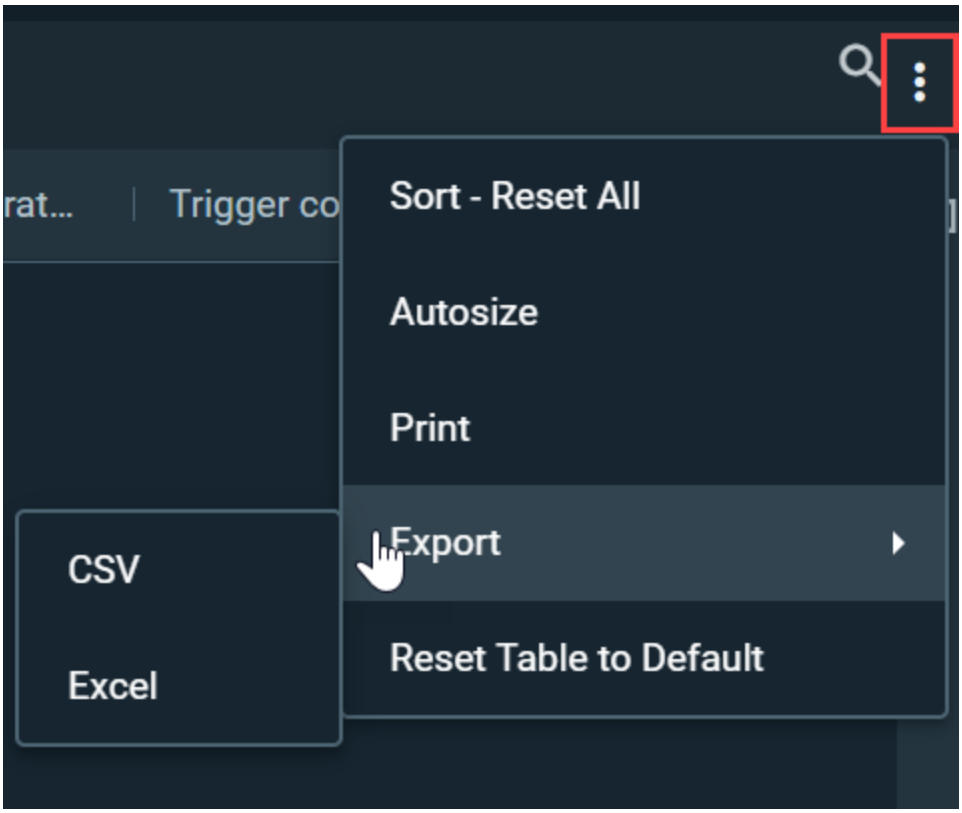
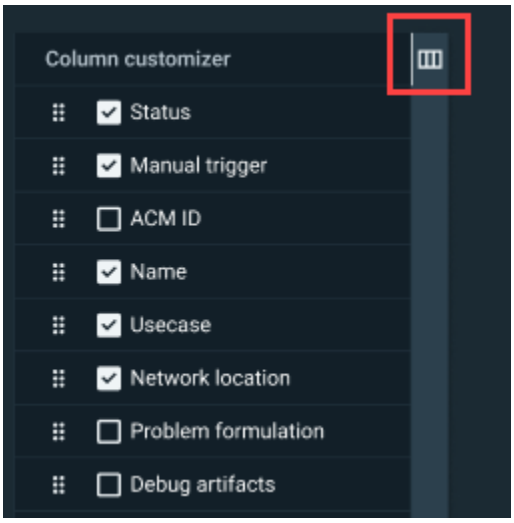


Figure 3. Table menu icon

Point to a column header, click the **Column customizer** icon, and select or clear columns, as desired.



5.2. Creating New Automation Configuration

To create a new configuration from the Automation configuration tab, click **new automation configuration**, a new tab opens. Complete the form fields for new configuration.

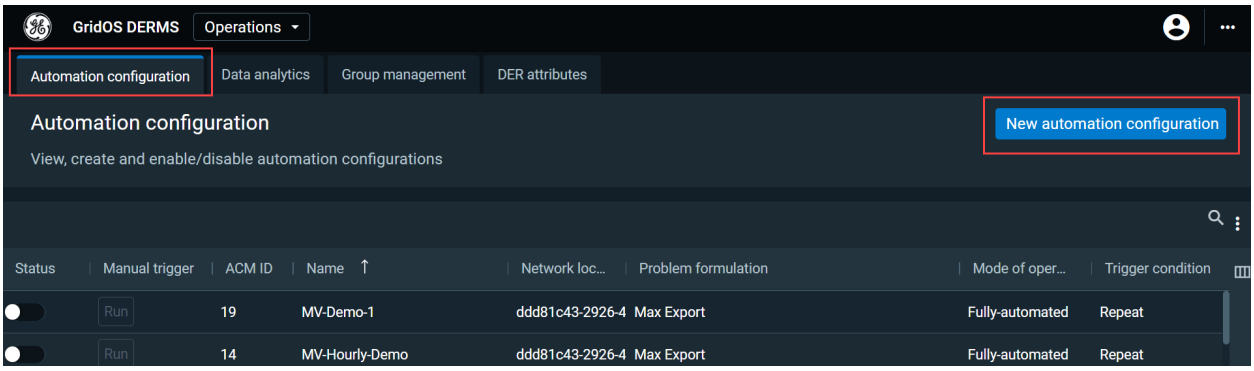
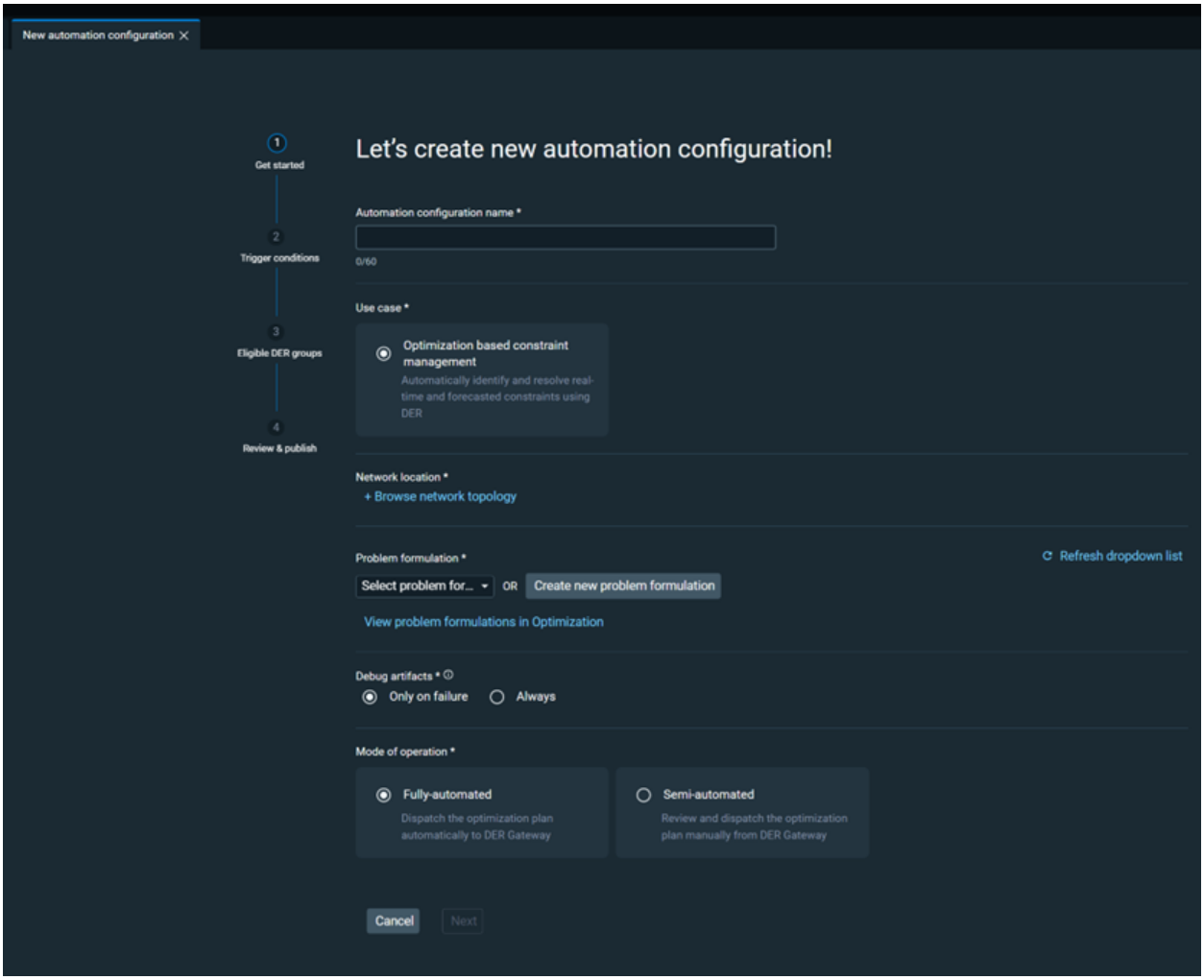


Figure 4. New automation configuration button

New Automation Configuration consists of 4 main steps:

1. Get Started
2. Trigger conditions
3. Eligible DER groups
4. Review & publish



5.2.1. Get Started (Create Automation Configuration)

To create an automation configuration.

1. Enter a unique name (maximum 60 characters). Use a naming convention that is meaningful to you.
2. By default, **Optimization based constraint management** is selected. This option automatically identifies and resolve real-time and forecasted constraints using DER.
3. Click **Browse network topology** to open the topology tree in the side panel. Network locations are listed with a checkbox next to their name, select the required checkbox.

> **Note**
>
> Maximum selection is limited to 1 substation only

4. Click on the drop-down list to select a problem formulation retrieved from Optimization.
5. To create a new problem formulation, click on **create new problem formulation** button and follow the steps creating_a_problem_formulation
6. **Debug artifacts**: Select the **Only on failure** radio button to generate debug artifacts when you created a trigger. You can download the report and debug artifacts from MinIO. These can aid in trigger troubleshooting, if needed.
7. **Mode of operation**: Select **Fully-automated** to dispatch the optimization plan automatically to DER Gateway. Select **Semi-automated** to review and dispatch the optimization plan automatically to DER Gateway.
8. Click **Next**.

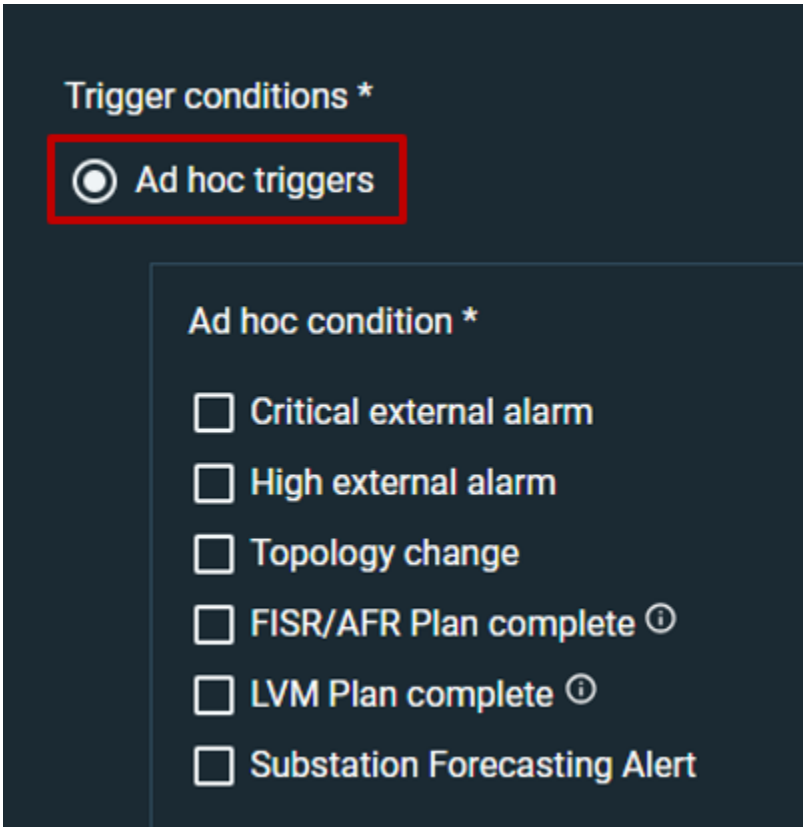
5.2.2. Configure Trigger Conditions

Ad-hoc Triggers

Ad-hoc Triggers enables users to send kafka messages to Operations. These kafka messages schema can be found in the reference file section of the user guide. Once a kafka message is received by operations. Operations will automatically trigger the corresponding automated constraint management configuration and queue it for optimization.

1. Select the Ad-hoc triggers radio button. For Ad-hoc condition, select the required checkbox.
2. Critical external alarm – User defined critical alarm
3. High External Alarm - User defined critical alarm
4. Topology change – Any changes to topology for the configured ACM substation
5. FISR (fault isolation service restoration), AFR (automated feeder reconfiguration) plan complete – If GE ADMS executed an FISR/ AFR plan, the completion status can be used to trigger a DER optimization re-computation

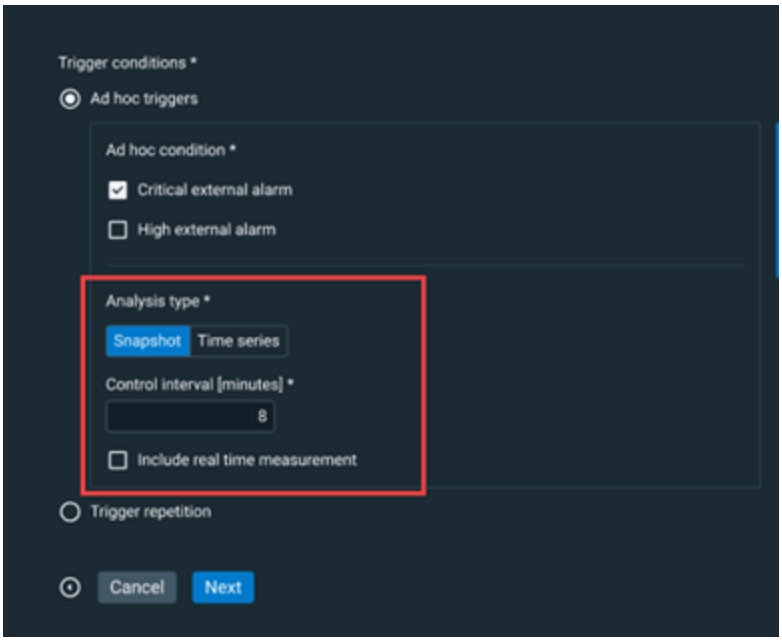
6. LVM – Load and Voltage/VAr management plan complete
7. Substation Forecasting alert – Any forecasting alert received from forecasting application



Analysis Type

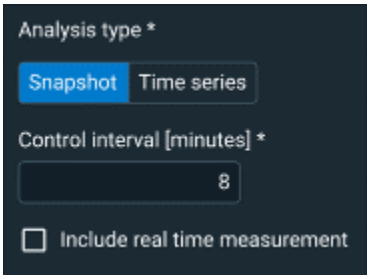
You can use the toggle option to select **Snapshot** or **Timeseries**.

> **Note**
>
> By default, Snapshot option is selected.



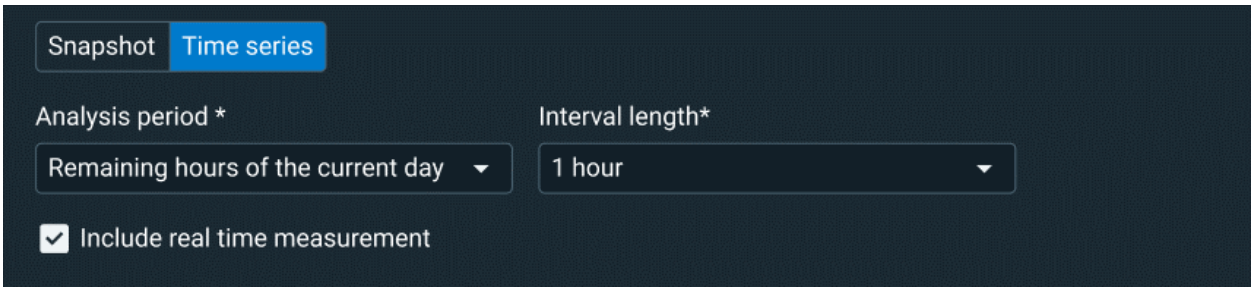
Snapshot

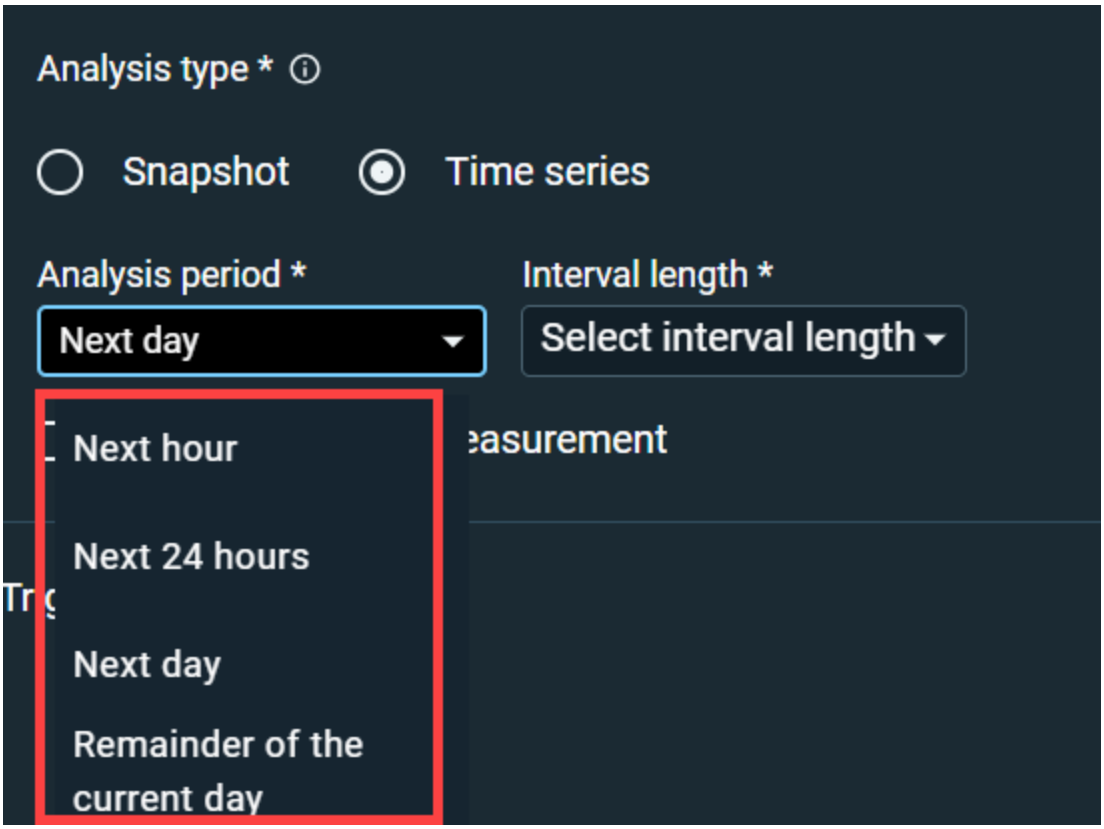
Select **Snapshot** option to perform a single interval analysis using the latest schedule data available. Enter the number of minutes in the **control interval duration** field and select the include **real time measurement checkbox**.



Time Series

Select **Time series** option to perform a multi-interval analysis by selecting the desired analysis period and interval length.



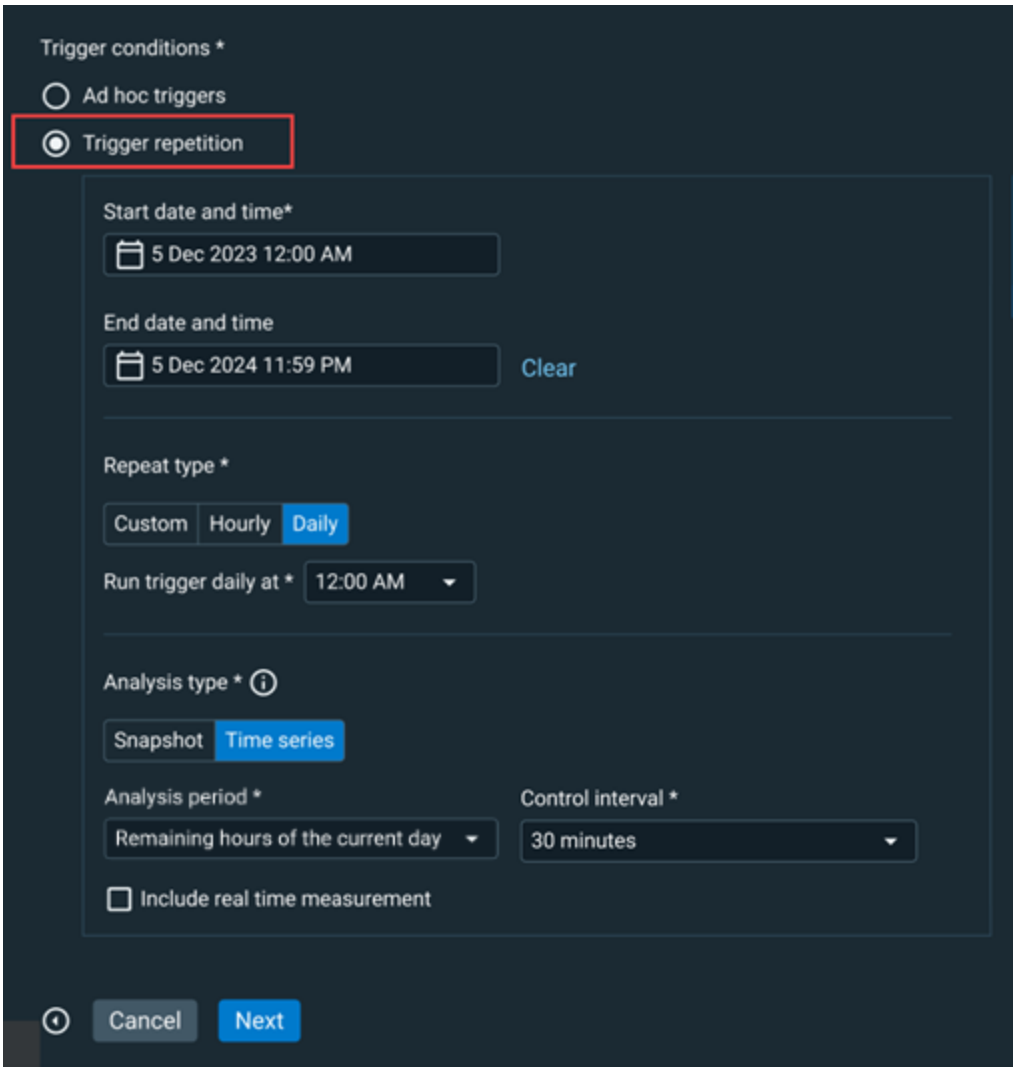


Analysis period: Select the desired duration from the drop-down list.

Interval Length: Select the desired time horizon.

1. Next Hour - This option uses the next hour's (60-minute period) data in the optimization calculation and analysis. For example, selecting a Next Hour time horizon means that schedule data forecasted for the next hour's 60-minute period (:00-:59) will be used in the analysis.
2. Next 24 hours - This option uses a 24-hour-period of data, starting at the beginning of the next whole hour in the optimization calculation and analysis. For example, selecting the Next 24 hours time horizon means that schedule data forecasted for the next 24 hours, starting the next hour (:00), will be used in the analysis.
3. Next day (24 hours) - This option uses the next day's data (24-hour period) in the optimization calculation and analysis, beginning at midnight (00:00-23:59).

Trigger Repetition



1. Select the calendar icon to choose the start and end date and time, this field Start date and time is mandatory.

> **Note**
>
> End date & time is not mandatory, always validate if the end date-time is always greater than start date-time.

2. Repeat type option, custom is opted as default. Select Hourly or Daily option and choose the minutes from the drop-down list to run the trigger.

The following table describes the resulting scenarios for the selecting options:

Snapshot

Custom	Custom – without end date	Runs a snapshot analysis every [x] minutes starting from [start date and time] indefinitely.

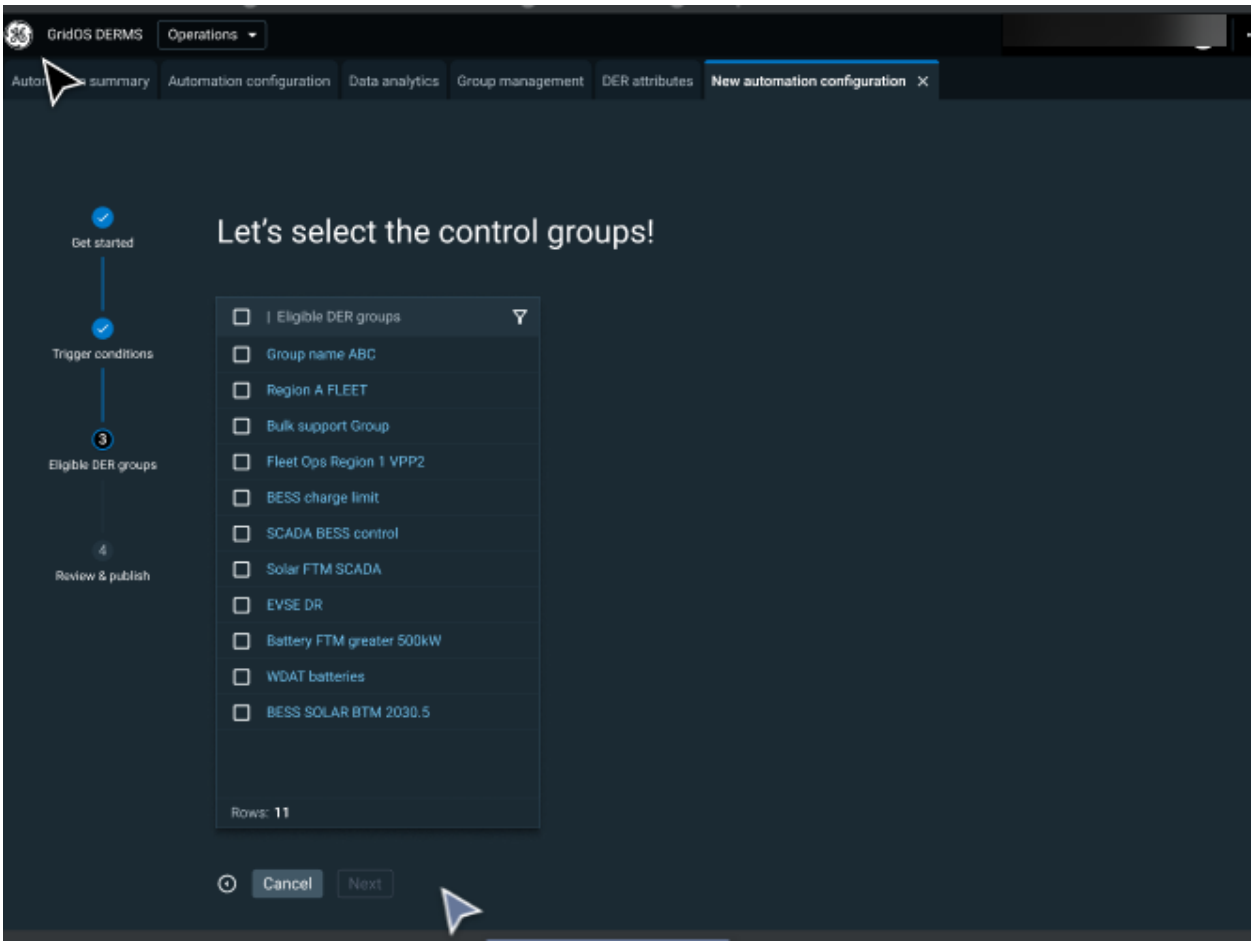
	Custom – with end date	Runs a snapshot analysis every [x] minutes starting from [start date and time] to [end date and time].
Hourly	Hourly – without end date	Runs a snapshot analysis every hour at [x] minutes past the hour starting from [start date and time] indefinitely.
	Hourly – with end date	Runs a snapshot analysis every hour at [x] minutes past the hour starting from [start date and time] to [end date and time].
Daily	Daily – without end date	Runs a snapshot analysis every day at [h:mm tt] starting from [start date and time] indefinitely.
	Daily – with end date	Runs a snapshot analysis every day at [h:mm tt] starting from [start date and time] to [end date and time].

Timeseries

Custom	Custom – without end date	Runs a time series analysis every [x] minutes starting from [start date and time] indefinitely.
	Custom – with end date	Runs a time series analysis every [x] minutes starting from [start date and time] to [end date and time].
Hourly	Hourly – without end date	Runs a time series analysis every hour at [x] minutes past the hour starting from [start date and time] indefinitely.
	Hourly – with end date	Runs a time series analysis every hour at [x] minutes past the hour starting from [start date and time] to [end date and time].
Daily	Daily – without end date	Runs a time series analysis every day at [h:mm tt] starting from [start date and time] indefinitely.
	Daily – with end date	Runs a time series analysis every day at [h:mm tt] starting from [start date and time] to [end date and time].

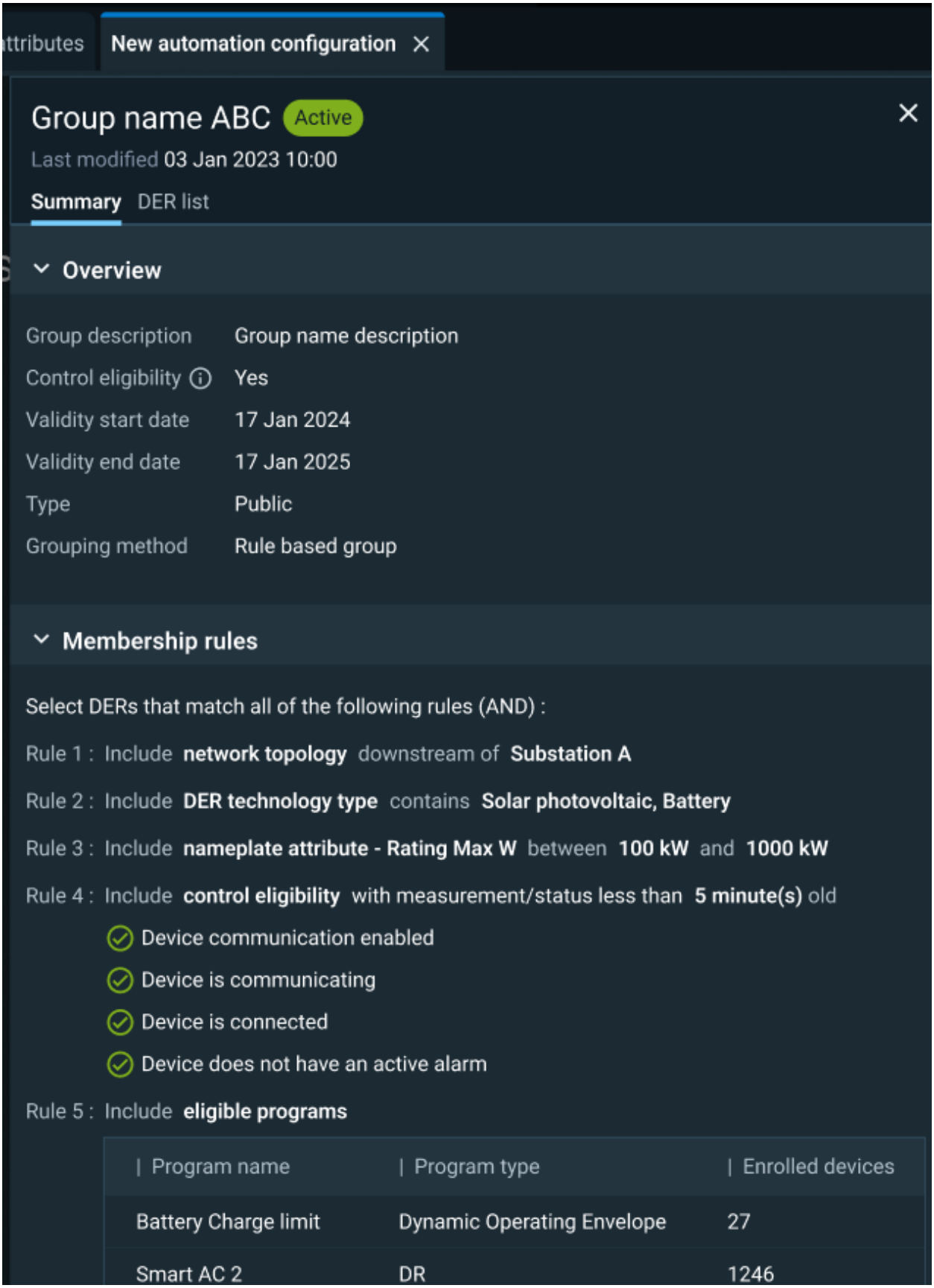
5.3. Eligible DER groups

1. You can select one or more eligible control groups for each automation configuration.
2. Click on the checkbox to select the required DER group.



Note

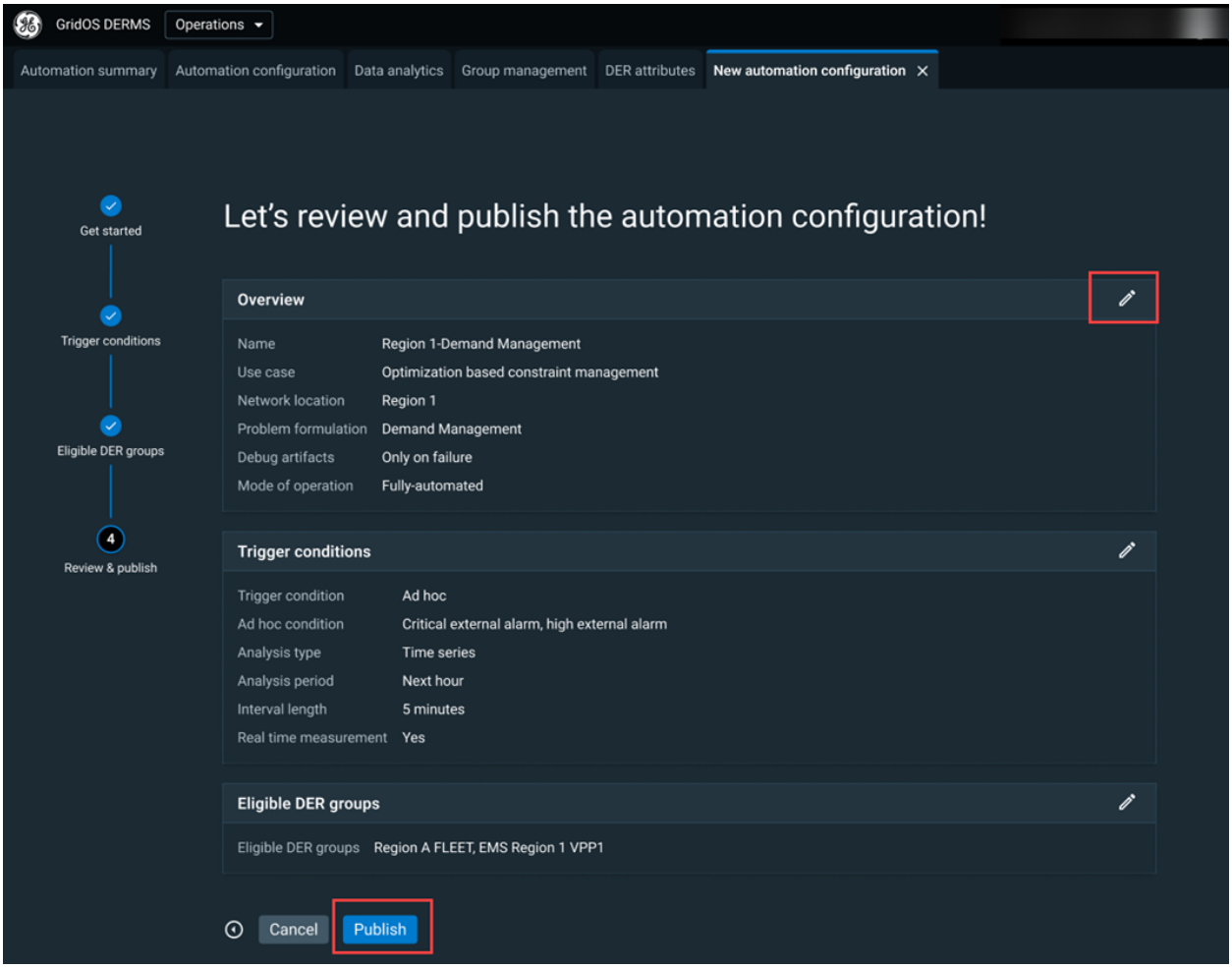
Click anywhere on the Eligible DER group, this opens a side panel with the group details.


ttributes
New automation configuration
Group name ABC
Active
Last modified 03 Jan 2023 10:00
Summary
DER list
Overview
Group description
Group name description
Control eligibility
Yes
Validity start date
17 Jan 2024
Validity end date
17 Jan 2025
Type
Public
Grouping method
Rule based group
Membership rules
Select DERs that match all of the following rules (AND) :
Rule 1 : Include network topology downstream of Substation A
Rule 2 : Include DER technology type contains Solar photovoltaic, Battery
Rule 3 : Include nameplate attribute - Rating Max W between 100 kW and 1000 kW
Rule 4 : Include control eligibility with measurement/status less than 5 minute(s) old
Device communication enabled
Device is communicating
Device is connected
Device does not have an active alarm
Rule 5 : Include eligible programs
Program name
Program type
Enrolled devices
Battery Charge limit
Dynamic Operating Envelope
27
Smart AC 2
DR
1246

5.4. Review & Publish

In the final step, you can view all the details.

1. Click **Edit** icon to make any changes to a required field.
2. Click **Publish**.



6. Data Analytics

In Data analytics, you can create and save any number of charts that compare data for assets on the network. The charts allow you to graphically plot the “near real-time measurements coming into DERMS.” The time horizon for a chart can be set to plus or minus 7 days from today.

The time horizon for the user is dynamic. This allows you to create a chart with a configurable time horizon. For example, if you select plus/minus 3 days, every time you load the chart, DERMS will plot past 2 days of data, today, and the next 2 days of data.

If you choose to compare assets, you can select up to 3 DER assets on the network for comparison of up to 5 parameters from their nameplate, measurement, or control parameter types.

If you choose to compare parameter types of a single DER asset on the network, you can select up to 5 parameters from up to 2 parameter types. Refer to Available parameters by parameter type.

Table 1. Available parameters by parameter type

Choose parameter type	To compare up to 5 parameters
Nameplate	Voltage Active Power Reactive Power Apparent Power Real Energy Rated Charging Active Power Rated Charging Apparent Power Rated Discharging Active Power Rated Discharging Apparent Power

Choose parameter type	To compare up to 5 parameters
Measurement	Current Voltage Active Power Frequency Reactive Power Apparent Power Apparent Energy Real Energy Reactive Energy Available Charge State of Charge Displacement Power Factor
Control	opModFixedPFAbsorbW (Fixed Power Factor Setpoint when absorbing Active Power) opModFixedPFInjectW (Fixed Power Factor Setpoint when injecting Active Power) opModFixedVar (Reactive Power Setpoint) opModFixedW (Charge / Discharge Setpoint) Reactive Power Active Power
Nameplate and Measurement	Voltage Active Power Reactive Power Apparent Power Real Energy
Nameplate and Control	Active Power Reactive Power
Measurement and Control	Active Power Reactive Power

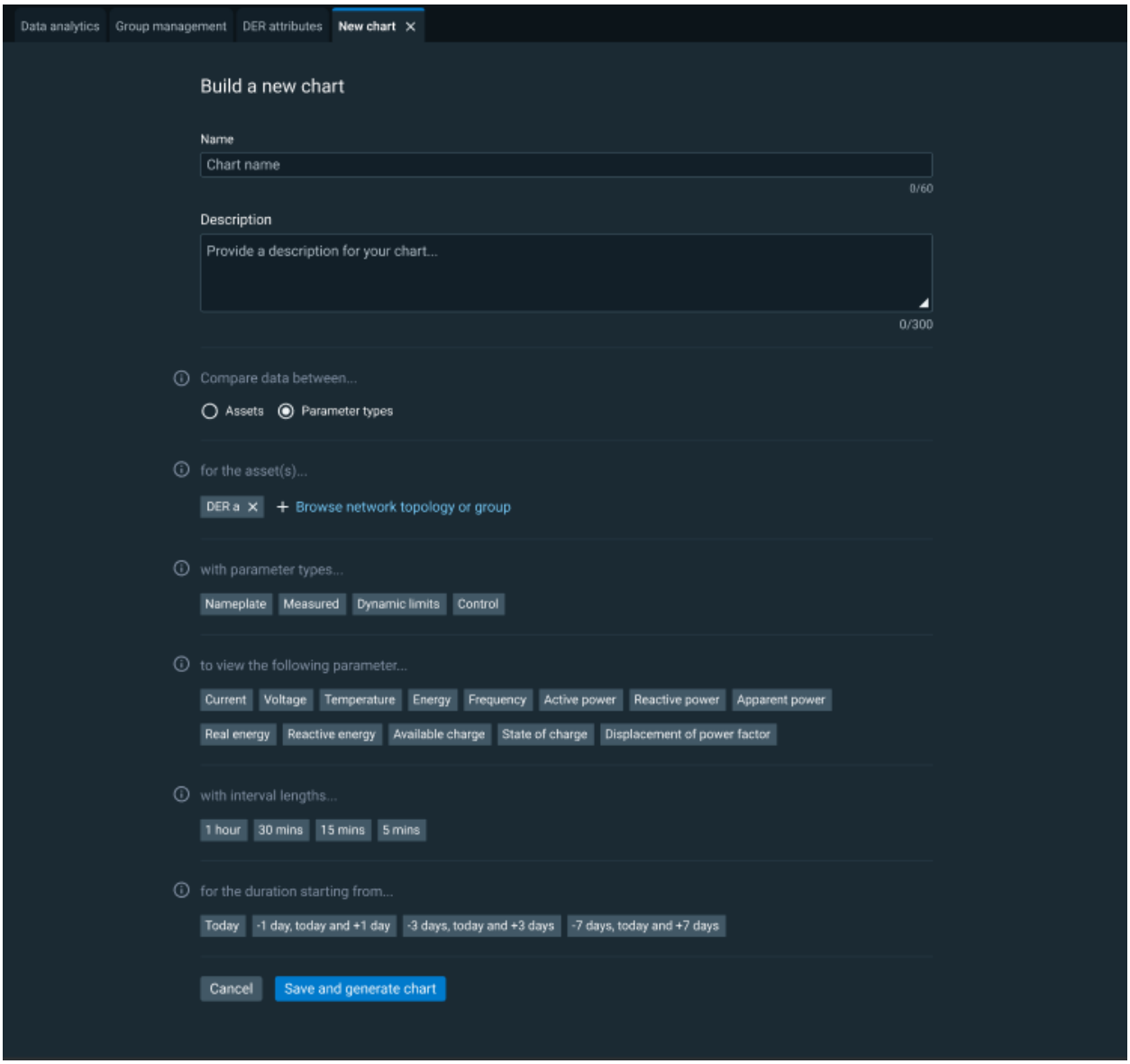
6.1. Creating a Chart

To create a chart from the Data analytics tab, click the Build a New Chart tile. A New chart tab opens. Complete the form fields.

1. **Chart name:** Give the new chart a unique name (maximum 60 characters). Use a naming convention that is meaningful to you.
2. **Chart description:** Optionally, enter a short description to describe the data sets and parameters being compared in this chart (maximum 300 characters).
3. Select a comparison attribute for the chart: Assets or Parameter types. Assets are individual DER assets, which you will select from the network topology tree. Parameter types are asset nameplates, measurements, or controls.
4. Select the DER assets you will include in the chart. Click **Browse network topology** to open the topology tree in the side panel. Individual DER assets are indicated with a checkbox next to their name. Select up to 3 assets if you are comparing assets, or 1 asset if you are comparing parameter types.
5. Select 1 parameter type as the subject of your chart: Nameplate, Measurement, or Control.
6. Select up to 5 parameters from the available options for comparison. See Available parameters by parameter type.
7. Select the time interval to display from the available options: 5 minutes, 15 minutes, 30 minutes, or 1 hour. The value displayed represents the average measurement for the interval.
8. Select the time duration for data to be included. This can be:
   a. One day (today)
   b. Three days (yesterday, today, and tomorrow)
   c. Seven days (today, the past 3 days and the next 3 days)
9. Click **Save** and generate chart. The new chart is saved to the Data analytics tab.

> **Note**
>
> If there is a data gap anywhere within the time horizon, the gap is not displayed on the chart. Data is interpolated for the time gap.

New chart form



6.2. Viewing the Chart Data

To view an existing chart and analysis, select a chart from the **Data analytics** tab. The chart opens in a new tab.

The chart's configuration is summarized in the chart header. The chart data is plotted for the configured time horizon and interval on the X axis, and the parameters on the Y axes.

If the chart configuration includes a Control parameter type, and the DER is enrolled in a program and is scheduled to be dispatched during the time horizon, a control event timeline overlay is displayed to show the upcoming control.

Chart Detail

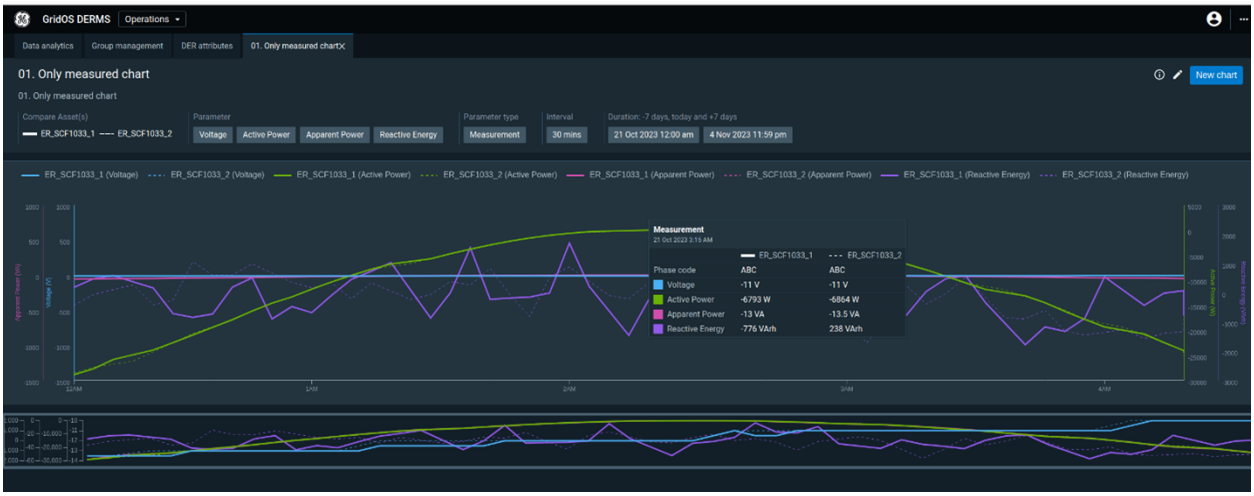
1. Chart header information. Displays the chart parameters as it was configured.
2. Control event overlay. Hover over this area to display the control event detail.
3. Control event detail. Control event ID, time, measurement, control value and program ID, for the DER enrolled in a program.

> **Note**
>
> Operations also stores a quality flag for each measurement, stored in DERMS. The chart will show a warning symbol ⚠ if the measurement data has a quality flag not equal to good.

4. Chart controls. Edit the chart or create a new chart.

For charts with a lengthy time horizon, you can zoom in to view a subset of the data to better understand data detail by selecting a segment in the timeline, below the chart. The selected segment is magnified in the main chart. Click **Reset Zoom** to zoom out to full scale.

Timeline segment magnification


GridOS DERMS
Operations
Data analytics
Group management
DER attributes
01. Only measured chart
01. Only measured chart
New chart
01. Only measured chart
Compare Asset(s)
ER_SCF1033_1
ER_SCF1033_2
Parameter
Voltage
Active Power
Apparent Power
Reactive Energy
Parameter type
Measurement
Interval
30 mins
Duration: -7 days, today and +7 days
21 Oct 2023 12:00 am
4 Nov 2023 11:59 pm
ER_SCF1033_1 (Voltage)
ER_SCF1033_2 (Voltage)
ER_SCF1033_1 (Active Power)
ER_SCF1033_2 (Active Power)
ER_SCF1033_1 (Apparent Power)
ER_SCF1033_2 (Apparent Power)
ER_SCF1033_1 (Reactive Energy)
ER_SCF1033_2 (Reactive Energy)
Measurement
21 Oct 2023 3:15 AM
ER_SCF1033_1
ER_SCF1033_2
Phase code ABC ABC
Voltage -11 V -11 V
Active Power -6793 W -6864 W
Apparent Power -13 VA -13.5 VA
Reactive Energy -776 VArh 238 VArh
12AM
1AM
2AM
3AM
4AM

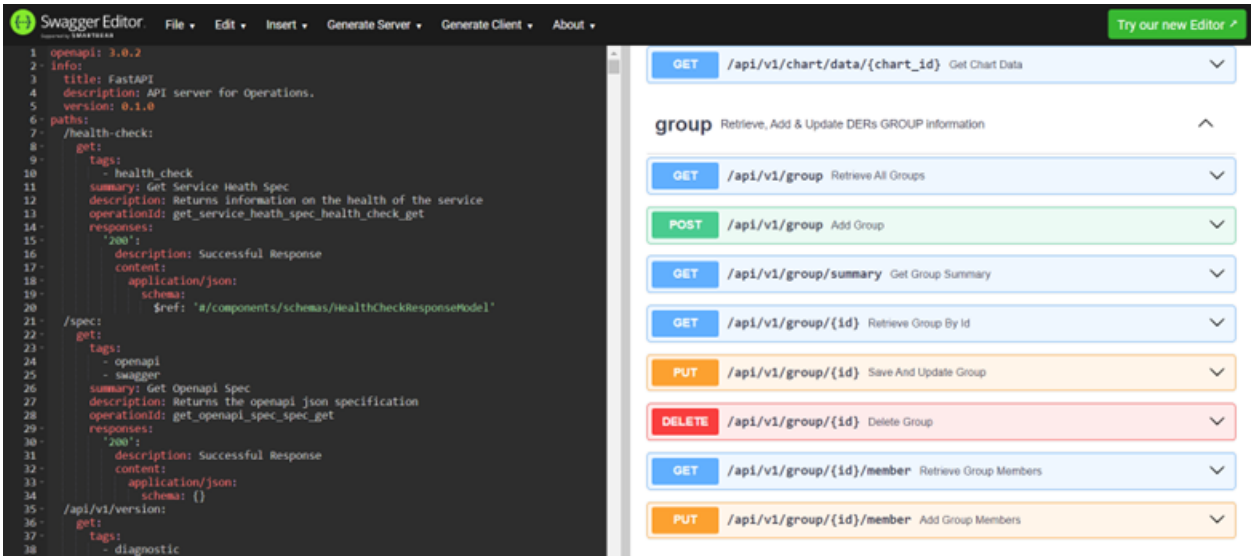
7. Group Management

Grouping of DER assets according to criteria you define can help you to view the combined, aggregated capabilities of the group, such as total rated capacity or total near real-time measurements of the group, for example. This simplifies aggregated information from numerous DERs into a single value for a quick snapshot of the grid.

Groups are created on the Group management tab in Operations. Creating a group requires that you define rules for DER membership in the group based on attributes, such as topology connections, DER type, program enrollment, or DER capabilities, such as rated max power, rated discharge power, rated voltage levels, etc. DERs must meet criteria defined in the rules to be included in a group. Once a group is created, DERs are added to the group automatically when they meet criteria for membership, and similarly, are dropped from the group if they no longer meet the criteria.

> **Note**
>
> Groups can be created via the user interface in Operations, or automatically via RESTful API. To obtain the API JSON file for group management actions, see https://<your derms url>/operations/api/v1/openapi.json. Copy and paste the JSON code into an API editor, such as Swagger Editor (https://editor.swagger.io/), for example. With this, you can retrieve, add, and update DER Group information via API.

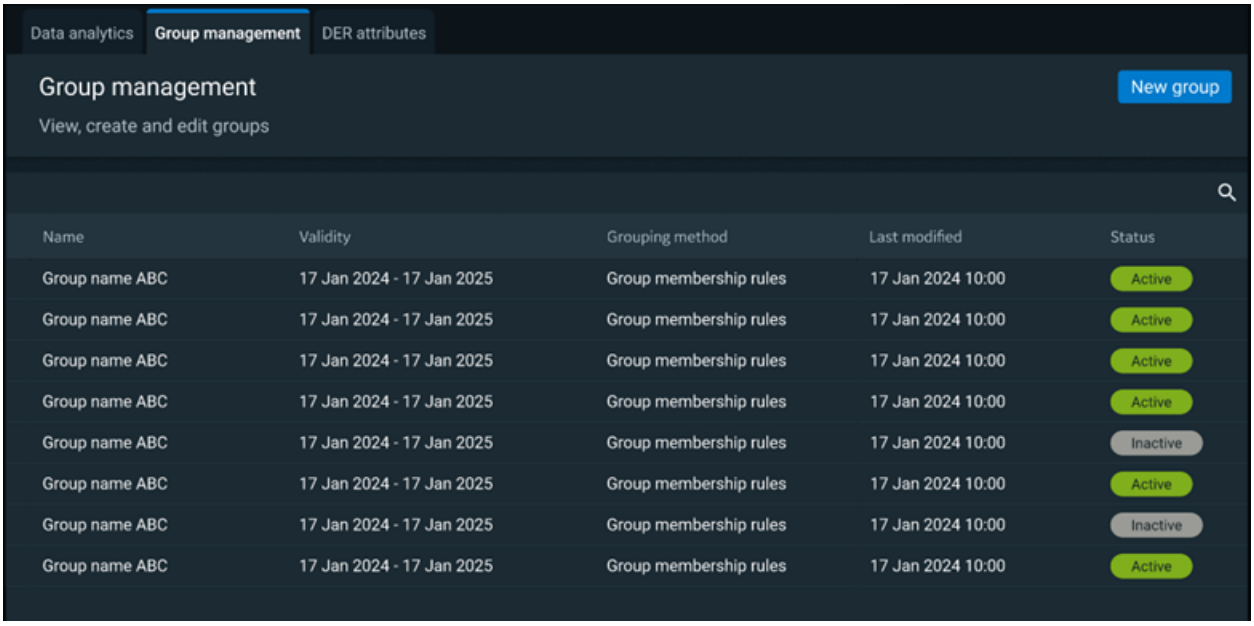
Swagger Editor



Existing groups can be accessed on the **Group management** tab in Operations. Click a group in the table to view or edit the details in a separate tab. The tab displays a summary of the group's configuration.

Groups can be in 1 of 3 statuses: Active, Inactive, or Updating. Groups are active when they are in the validity period. Groups are in an Updating status when DER data points are being actively received by the system. (Inactive groups are outside the validity period.)

Group Management tab



7.1. Creating a Group

To create a group from the Group management tab, click New group. A New group tab opens. Complete the form fields for group configuration.

1. **Name:** Give the new group a unique name (maximum 60 characters). Use a naming convention that is meaningful to you.
2. **Description:** Optionally, enter a short description to describe the group's validity and membership to help you recognize the group's purpose (maximum 300 characters).
3. **Validity:** Select a Start date and an optional End date to define the duration the group will be active. The validity period is inclusive of the start and end dates. This can be edited later to extend the validity period, if needed.

   > **Note**
   >
   > Leaving the End date blank means the group will remain active indefinitely.

4. **Grouping method**

   > **Note**
   >
   > In this release, a **Rule-based group** is the only grouping method available.

5. **Membership rules:** Use the dropdown selections to build 1 or more rules that specify the group membership criteria for DERs. Start by selecting the AND/OR function that will apply to the rules, then begin defining rules, one at a time.

6. When ready, click **Save changes.** The group appears on the **Group management** tab.

> **Tip**
>
> Membership rules use an AND/OR function to determine DER membership in the group. If you select **all the following rules (AND)**, only DER that meet the criteria in all the rules you define will be included in the group. If you select **any of the following rules (OR)**, DER that meet criteria in any 1 rule you define will be included in the group.
>
> Use the Include/Exclude parameter to specify what category of DER to include or exclude from group membership.
>
> Use the Category dropdown to define a membership rule with quantifiable limits to the category of DER.

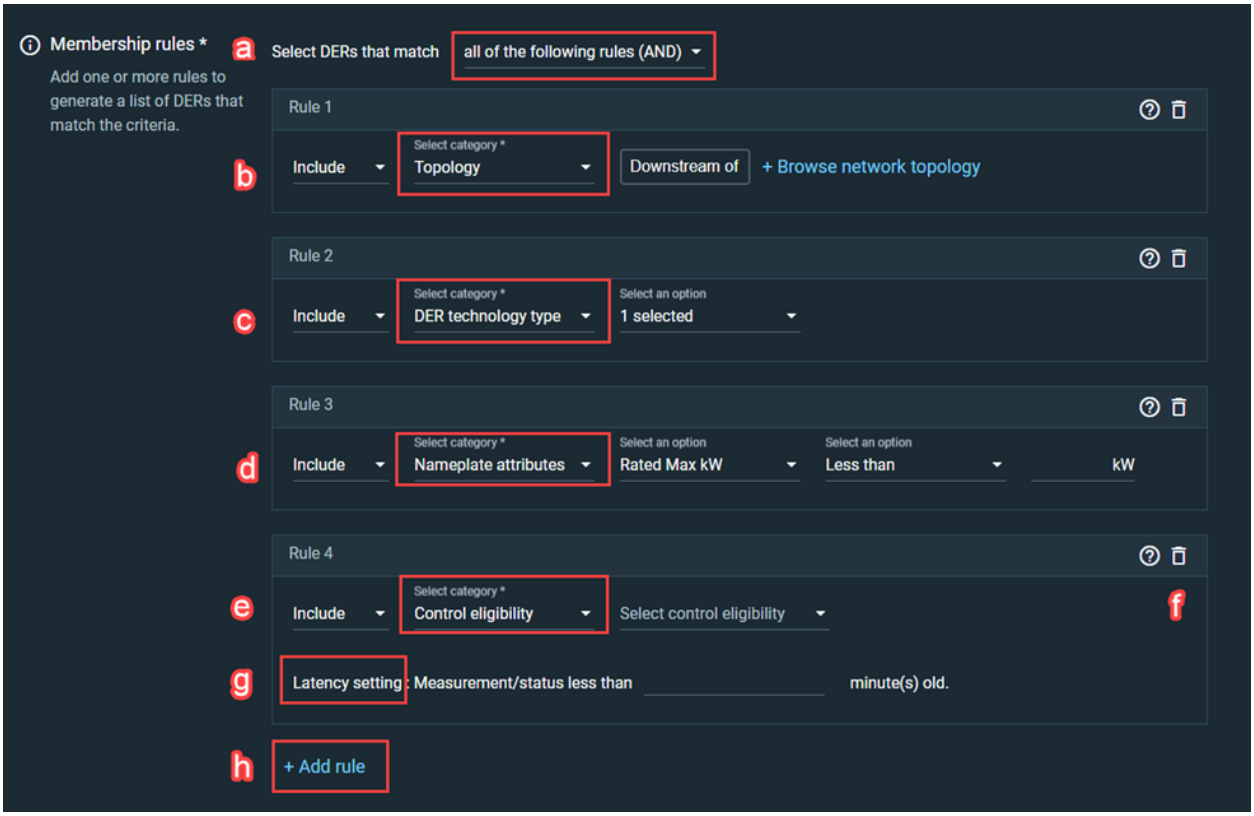


Figure 5. Defining Group Membership Rules

In this example of membership rule configuration, 3 rules are configured:

1. The DER must meet all the rules defined, otherwise they are not included in the group. This step uses the **AND** function.
2. Rule 1 states that only DERs in the network that are downstream from the specified substation are included in the group. Click **+Browse network topology** to select the node from the network topology tree. Any assets higher up in the topology are excluded from the group.
3. Rule 2 states that only DERs of a certain technology type are included in the group. Of the 3

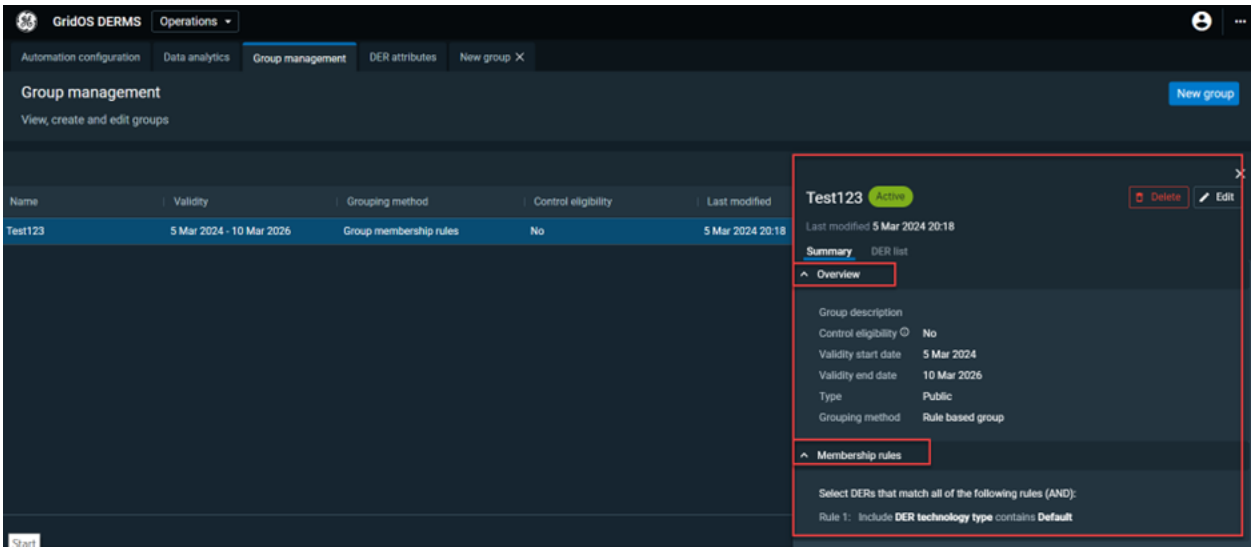
technology types to select from, only solar PV and batteries DER are included in the group. Other energy source DER are excluded.

4. Rule 3 states that only DER with a rated max energy between 100 kW and 1000 kW are included in the group. Any DER with rated max energy outside this range are excluded.
5. Rule 4 states that only DER that meets the control eligibility criteria are included in the group. Select the required control eligibility options from the dropdown list. In the Latency setting field, enter the digit 5 to define the duration.
6. Click the trash can icon to delete the rule. Deleting a rule does not affect the remaining rules.

> **Note**
>
> Latency setting: Allows you to define how old the measurement can be before considering the measurement to be out of bounds for group evaluation.

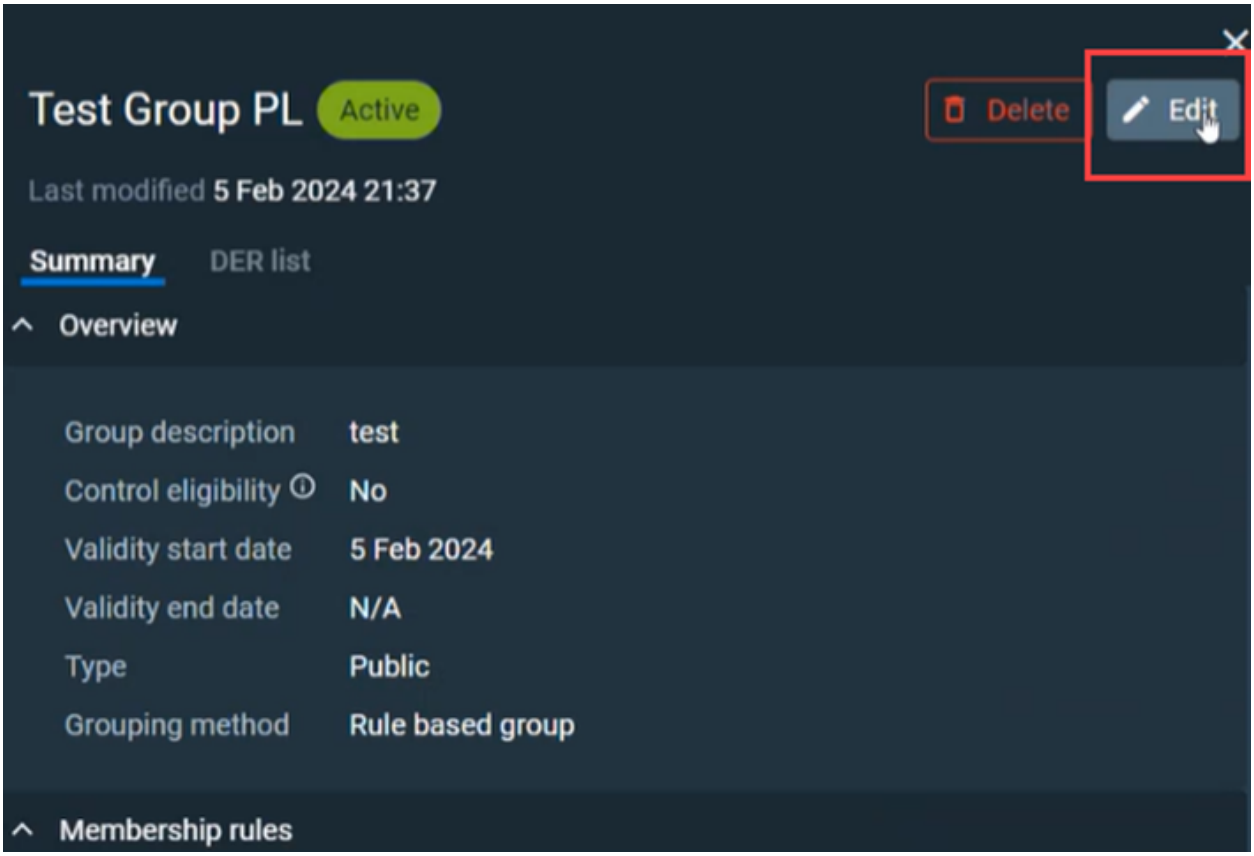
7. Enter the number of minute(s) in the field.
8. Click **+Add Rule** to continue adding rules until you have defined all criteria for DER inclusion in the group.
    - Selecting a row in the table displays the group's details in a right-hand side panel. On the Group summary tab of the panel, you can view the group's configuration.

a) You can now view the group's details in a side panel. Click on the row to view the side panel.

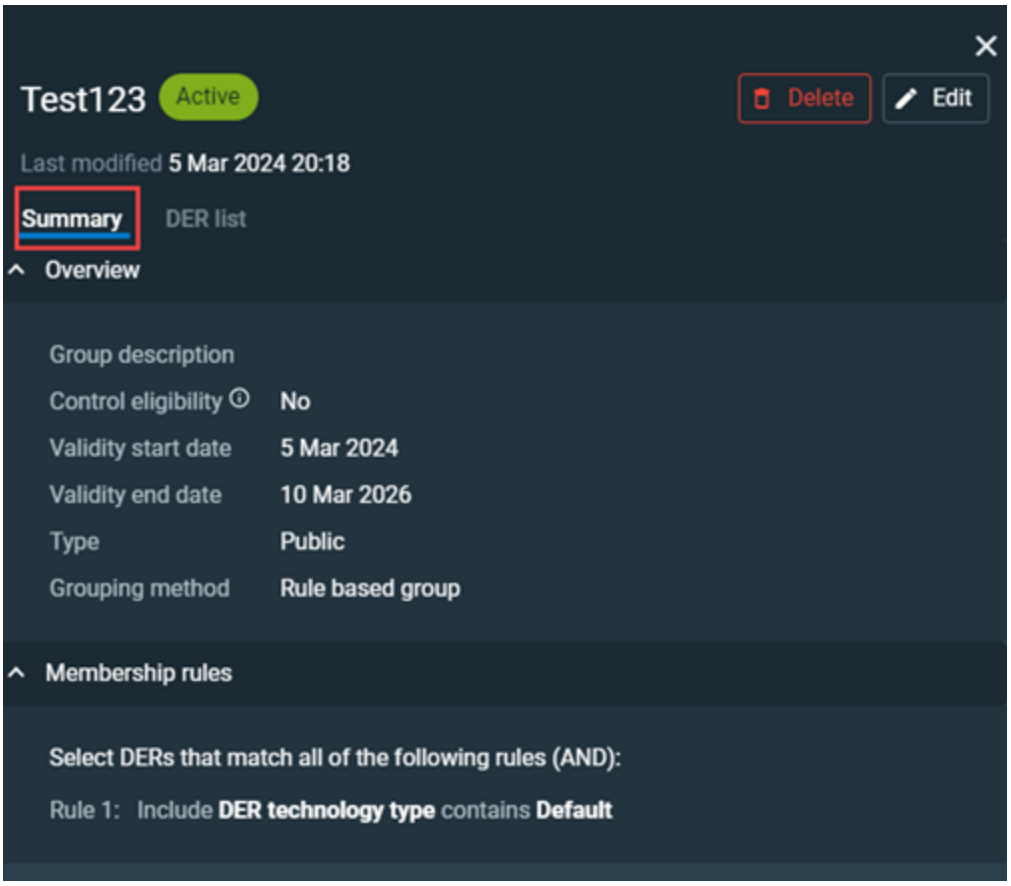


b) In the side panel window, you can **Delete** and **Edit** the group configurations.

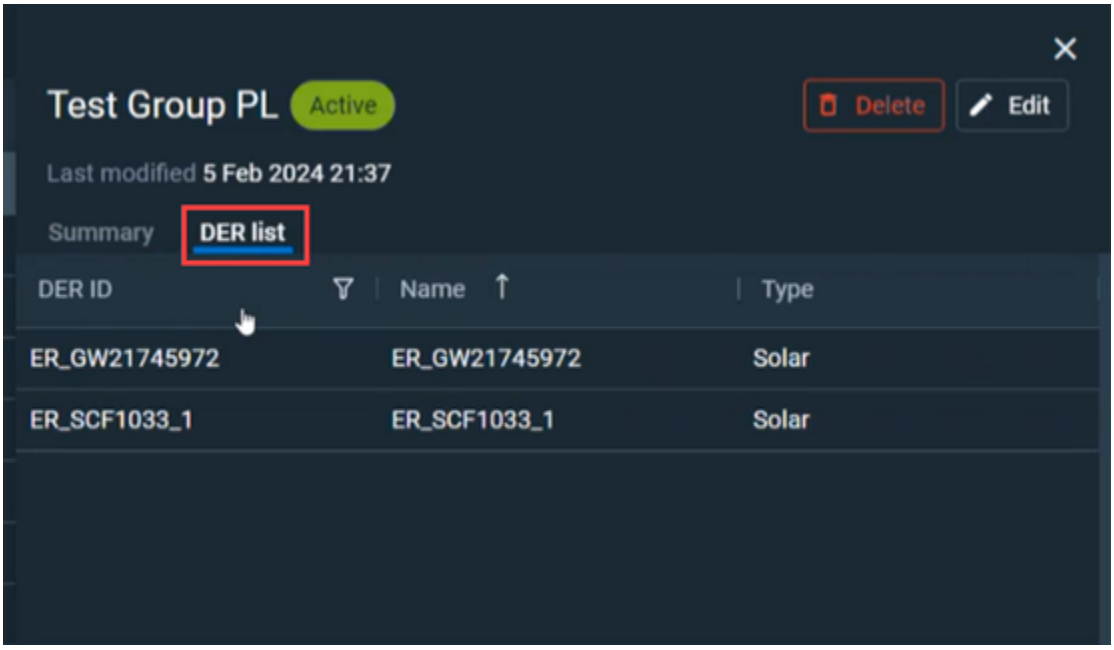
c) Click **Edit** and a new tab opens to input the required details.



d) Click **Summary** tab to view the **Overview** and **Membership rules** of the respective group.



e) Click **DER list** tab to the list of DERs assigned/tagged for the respective group.



7.2. Customizing the Group Management View

For enhanced accessibility, the table has the following options.

Refer to Customizing the Data Table View

You can drag the column pane to view or display the column names (attributes). By using this option, you can drag to reset, freeze columns, sort multiple columns, and organize the table view.

> **Note**
>
> The first two columns are fixed, you can only use the drag and view option from the third column.

8. DER Attributes

You can view the DER-specific attributes of a group in the **DER Attributes** area of the module. A DER asset must have membership in a group to be able to see this. For a given asset, you can view these attributes:

Categories: DER Nameplate attributes (anything rated), DER metadata (name, type, etc.), DER measurements (measured quantities), DER status information (connection status, etc.).

8.1. Viewing DER Attributes

From the DER attributes tab, search or select a group. Details are displayed in the side panel. You can customize the columns to include or exclude attributes in view or organize the columns in a different order. Always reload the page to get the latest values.

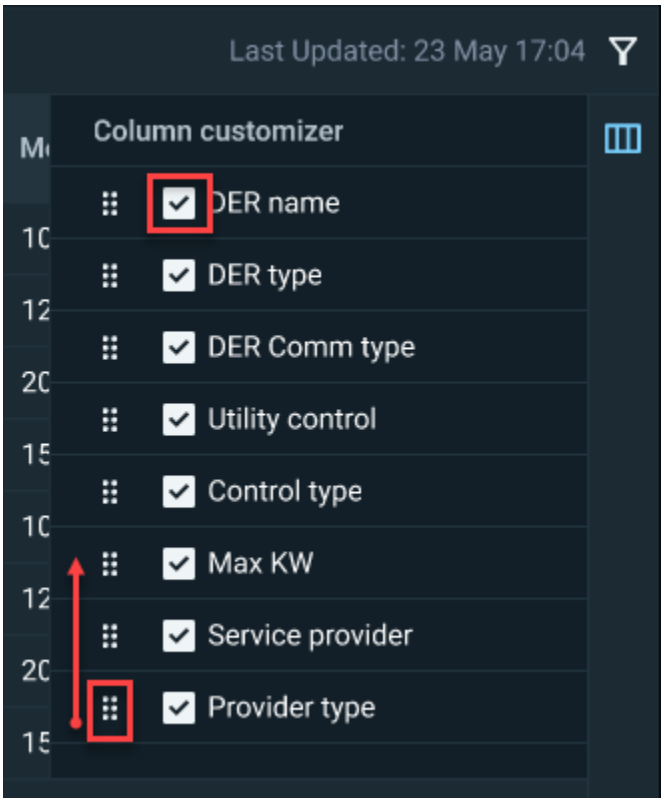
8.1.1. Customizing the Attributes Table View

For enhanced accessibility, the table has the following options.

Refer to Customizing the Data Table View

1. To customize the columns in the attributes table, click the column customizer icon from the top right of the side panel.
2. Select or clear the checkbox next to the column names you want to show or hide in the attributes table.
3. Click and drag the column organizer to move a column to a new position in the attributes table.

Column customizer



8.1.2. Searching for Groups or Individual DER

To search or filter for a group name or a DER name:

1. Click the filter icon from the top right of either the Groups panel (to search for a group) or the detail side panel (to search for a DER name).
2. From the shortcut menu, select a filtering criterion.
3. In the Filter text box, enter a keyword to search. The table is filtered and displays records containing the keyword. Combine filtering criteria to narrow the search results.

8.2. Visual Representation of DER Settings

Distributed Energy Resources (DERs) vary widely in type and configuration. Due to their complexity and the diverse requirements of different grid locations, DERs support a range of settings tailored to the specific needs of each utility or even individual substations.

8.2.1. View DER Settings (Curves)

You can now view both active and supported curves for DERs to enable further analysis, such as measurement and verification.

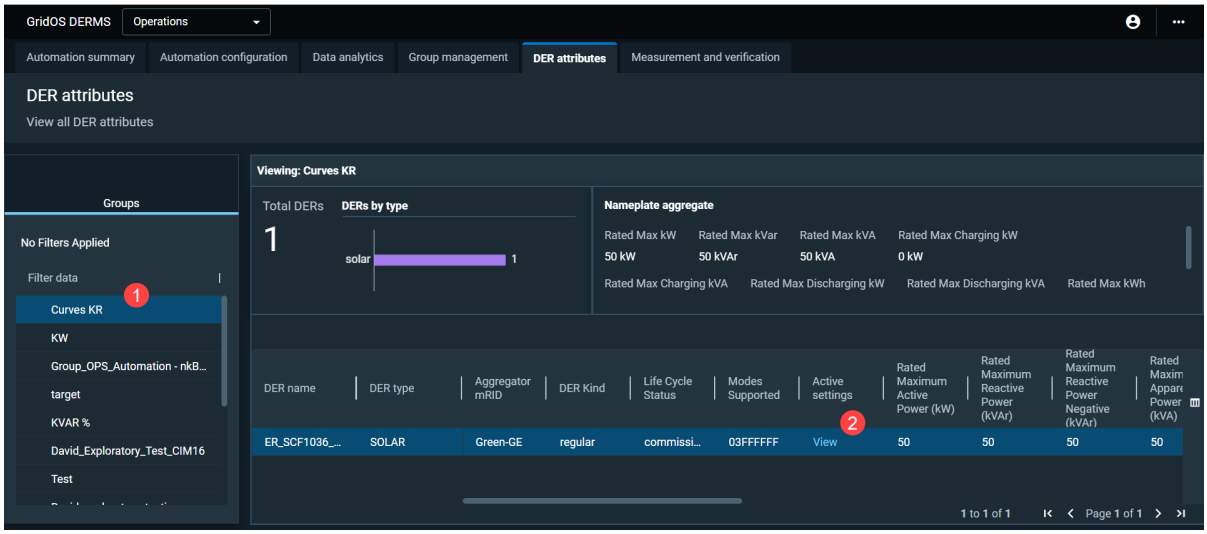
This feature aims to enhance the DERMS system by enabling users to:

- View detailed settings and configurations for each DER.
- Edit and manage these configurations as needed.
- Summarize settings across DERs for group-level insights.
- Bulk update configurations with support for automation.

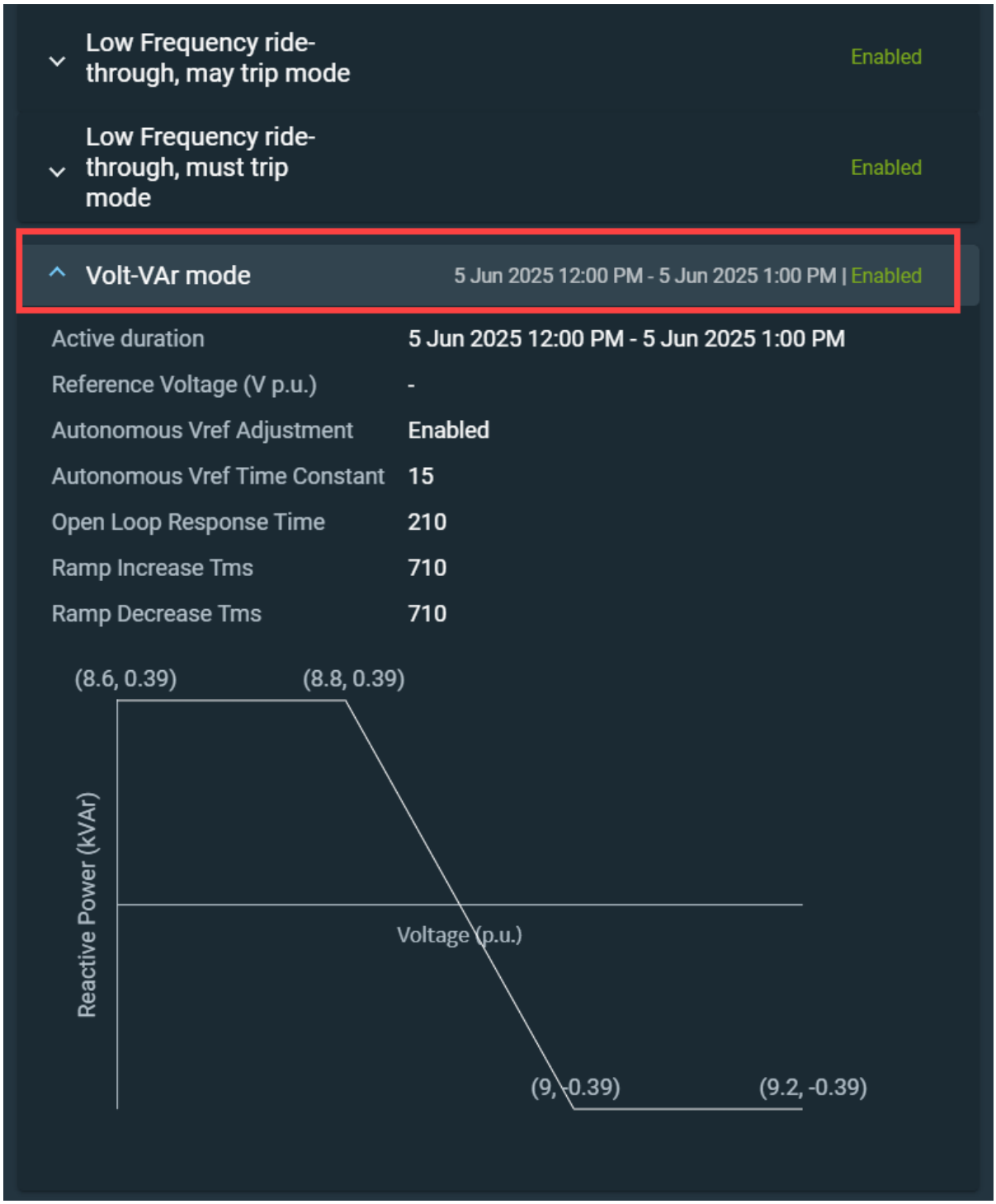
These capabilities ensure that DERMS users can effectively manage DER configurations, which are often unique and dynamic, to maintain grid reliability and performance.

To view the applied settings for a DER group:

1. In the **DER Attributes** tab, select the desired group from the list in the right-hand pane.
2. In the **Active Settings** column, click the **View** hyperlink next to the group.
3. A separate pop-up window will appear, displaying the associated curves.



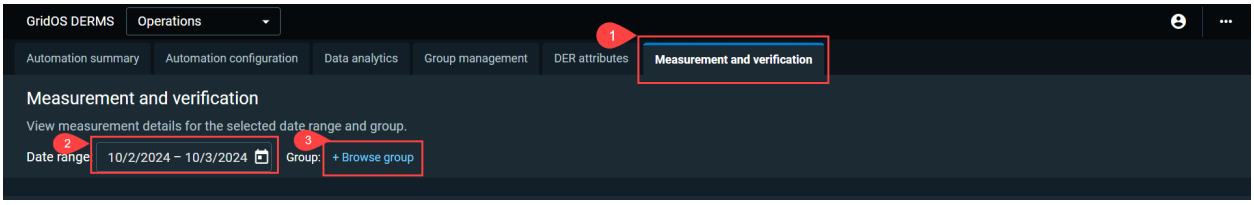
4. Click the arrow icon to expand and view the curve details.


Low Frequency ride-through, may trip mode
Enabled
Low Frequency ride-through, must trip mode
Enabled
Volt-VAr mode
5 Jun 2025 12:00 PM - 5 Jun 2025 1:00 PM | Enabled
Active duration
5 Jun 2025 12:00 PM - 5 Jun 2025 1:00 PM
Reference Voltage (V p.u.)
-
Autonomous Vref Adjustment
Enabled
Autonomous Vref Time Constant
15
Open Loop Response Time
210
Ramp Increase Tms
710
Ramp Decrease Tms
710
(8.6, 0.39)
(8.8, 0.39)
Reactive Power (kVAr)
Voltage (p.u.)
(9, -0.39)
(9.2, -0.39)

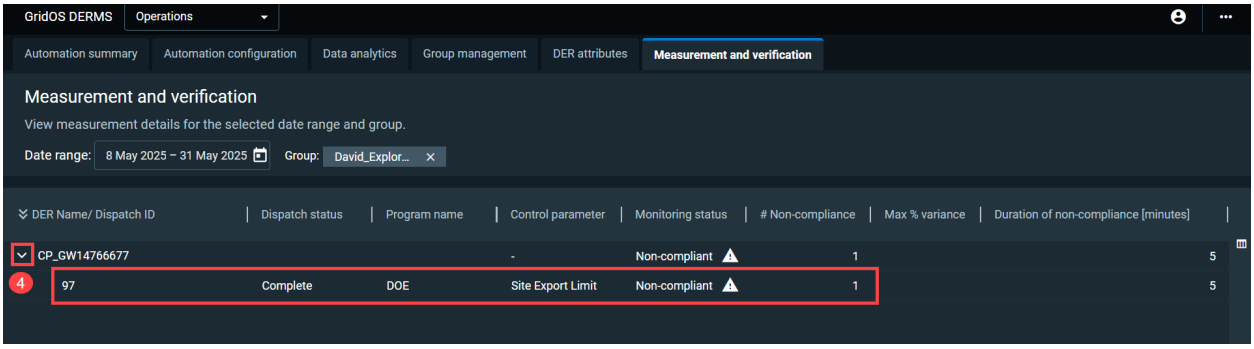
9. Measurement and Verification

In this tab, you have the ability to evaluate a data-set based on a measurement and verification rule or configuration. This service can be accessed by other DERMS applications for measurement and verification computation.

9.1. View Measurement Details



1. Click on the Measurement and Verification tab.
2. Select required date range by clicking the calendar icon.
3. Click on the **+Browse group** icon to choose the required group.
4. Click on the arrow next to the DER ID to view the summary of the corresponding ID.



9.2. Computing Compliance for an Individual Measurement

How to determine compliance or non-compliance for an individual measurement?

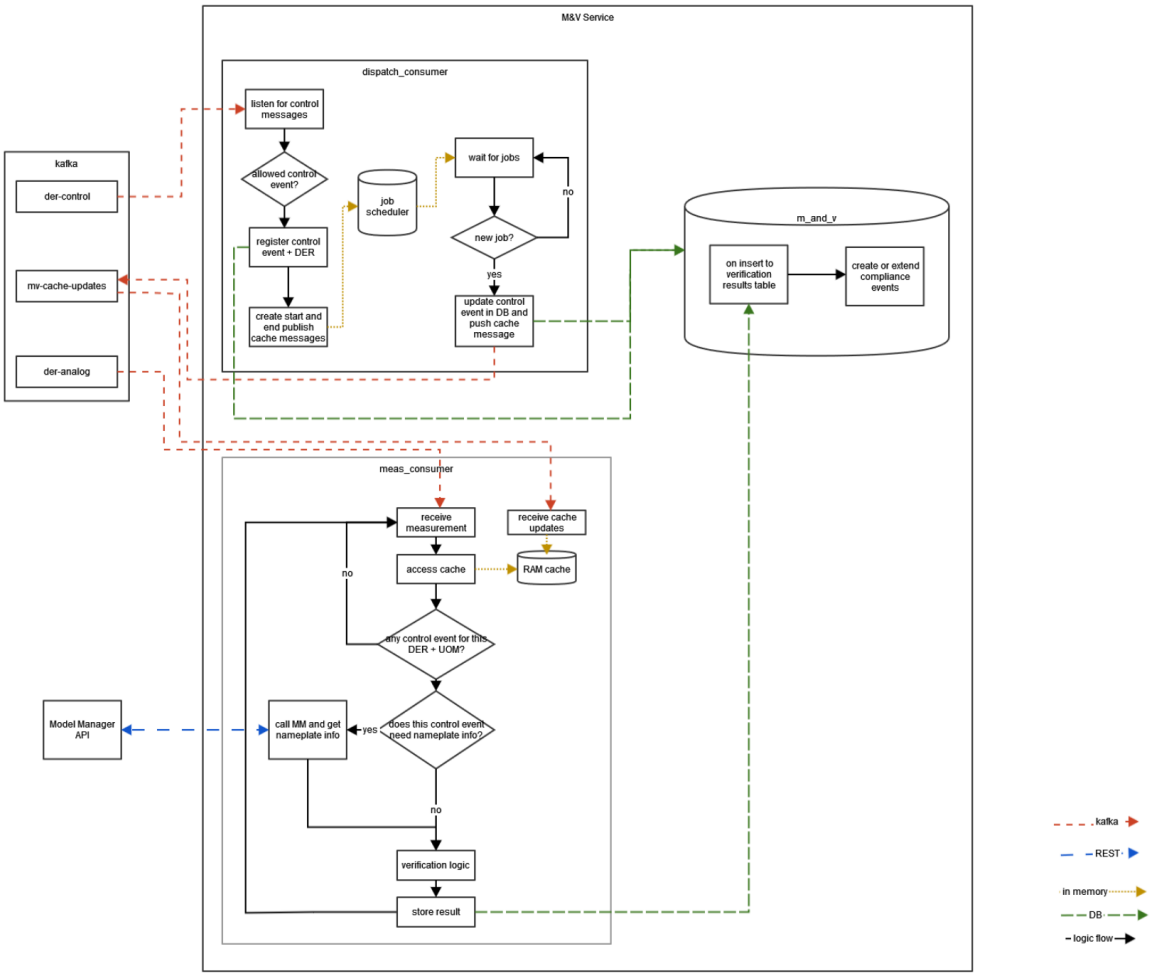
This algorithm only handles one type of control event, the type that specifies a specific target value for all DERs in the event (e.g. 1500 kW, 500 kVAR, etc).

This algorithm has a special case to handle a control event with a target value of 0, because without this special case the math would not work out correctly. This special case will require a configuration value set at deploy time.

This configuration value could be called something like *EVENT_TARGET_OFFSET*, and should default to 5000 (in base units, applicable to all metric types).

- When a measurement is received, it undergoes evaluation for compliance or noncompliance, and the result is stored as a measurement evaluation.
- A compliance event is a chronological series of measurement evaluations that all have the same compliance status, and belong to the same DER and control event in other words, for each DER and control event:
  - A compliance event begins when a measurement evaluation is stored with a compliance status that is not the same as the previous measurement evaluation.
  - A compliance event continues when a measurement evaluation is stored with a status that is the same as the previous measurement evaluation.
  - A compliance event ends when the current measurement evaluation is different from the last-evaluated measurement evaluation. There is no assumption about the compliance status between compliance events, a compliance event with only one measurement evaluation will have a duration of 0.

> **Note**
>
> The raw results for Measurement and Verification (M&V) are available on the designated Kafka topic. For detailed setup and access instructions, please refer to the Installation Guide.



The below *figure 1* shows compliance is calculated normally without making any assumptions.

The below *figure 2* shows how compliance is calculated in cases where an assumption is made based on measurements.

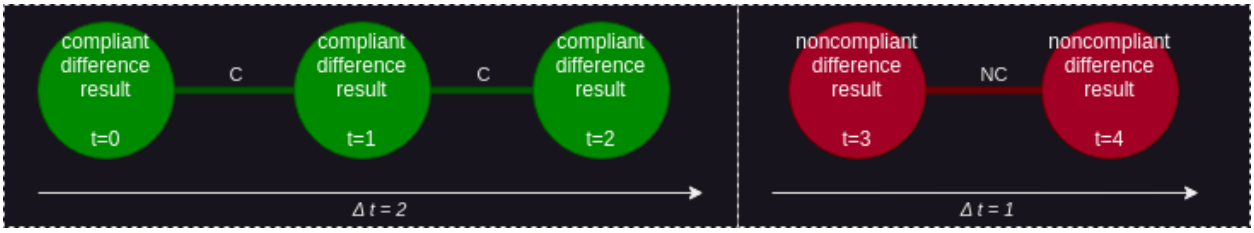


Figure 6. Compliance calculated without assumptions

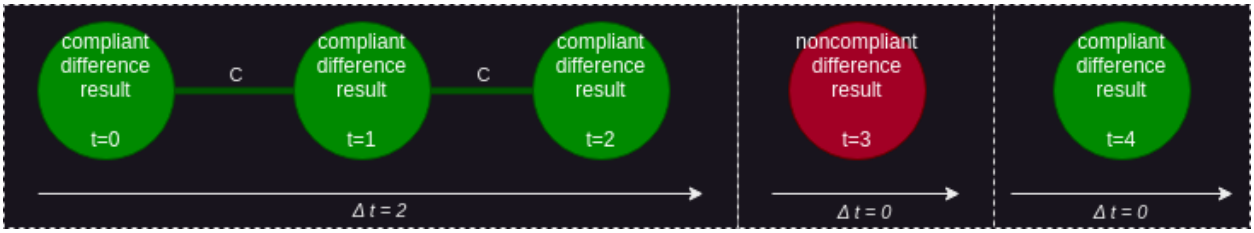


Figure 7. Compliance calculated based on measurements

10. Feature Flags

This chapter lists the Feature Flags available on the Operation module. Enabling feature flags has a direct dependency on the modules that are deployed with Operations, see matrix below for the Operations features that would be enabled for the feature flag.

> *When Optimization is deployed without Program Manager*
>
> **Note**
>
> - You cannot create automation configuration. This requires a control eligible group which needs at least one program.
> - When you create an automation configuration the list of groups will only show control eligible groups. In this scenario, as Program Manager would be turned off, this does not allow you to create a control eligible group.

10.1. Enabling Feature Flags

In the deployment manifests the following values need to be enabled. In addition the module intending to be enabled needs to be deployed on your environment.

```
featureFlags:
  optimization: false
  programManager: false
```

> **Note**
>
> Enabling/Disabling feature flags currently does not support the Backend APIs.

10.2. Feature Matrix

	Optimization	Program Manager
Automation Summary	Yes	No
Automation Configuration	Yes	No
Program Rules	No	Yes
Remote Control Eligibility with Program Evaluation	No	Yes

11. OpenStreetMap PBF Import for Air-Gapped Deployments

This document provides instructions for importing OpenStreetMap PBF files to enable the Geoview feature in air-gapped environments.

11.1. Overview

To enable the Geoview feature in air-gapped deployments, you have two primary options:

11.1.1. Option 1: Connect to Self-Hosted Map Server (Recommended)

If our customer has a self-hosted ArcGIS or OpenStreetMap server that supports the XYZ tile standard, you can configure the Geoview feature to connect directly to their existing infrastructure. This is often the more common and preferred approach as it:

- Leverages existing customer infrastructure
- Requires minimal setup and maintenance
- Provides access to customer-specific map data and styling
- Eliminates the need for data import and storage

Any map server that serves tiles using the XYZ tile standard (e.g., `https://mapserver.example.com/{z}/{x}/{y}.png`) can be integrated with the network viewer.

11.1.2. Option 2: Import OpenStreetMap Data Locally

For deployments without access to compatible external map servers, you can import OpenStreetMap data into a local PostGIS database. This process involves:

1. Download a PBF file for the desired region from https://download.openstreetmap.fr/
2. Place the PBF file on a Kubernetes node or Docker environment in an Internet-connected environment
3. Create a temporary Kubernetes pod or Docker container to import the PBF file into a PostGIS database
4. Export the Kubernetes volume and transfer it to the target air-gapped Kubernetes environment
5. Deploy Operations with the OpenStreetMap tile server enabled

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11.2. Prerequisites

- Access to https://download.openstreetmap.fr/ for downloading PBF files
- Docker image `overv/openstreetmap-tile-server:2.3.0`
- Kubernetes cluster or Docker environment for performing the import
- Sufficient storage space (imported database is 10-100 times larger than the PBF file)

11.3. Step 1: Download OpenStreetMap Data

11.3.1. Standard Downloads

Download the OpenStreetMap data matching your project area from https://download.openstreetmap.fr/.

For middle-sized countries, PBF files are typically 50-200 MB (e.g., `saudi_arabia.osm.pbf`, `benin.osm.pbf`).

11.3.2. Handling Large Areas

For larger areas with multi-GB PBF files that may be difficult to download or contain unnecessary data, use osmconvert to merge country-specific files.

Example: Merging Togo and Benin into a single file

```
osmconvert64-0.8.8p.exe benin.osm.pbf --out-o5m | osmconvert64-0.8.8p.exe - togo.osm.pbf -o=BT.osm.pbf
```

> **Warning**
>
> Incremental import is not supported. All OSM data must be available in a single PBF file as `openstreetmap-tile-server` uses `osm2pgsql --create`.

11.4. Step 2: Connecting to Self-Hosted Map Servers

Before proceeding with local data import, consider connecting to your customer's existing map infrastructure if available.

11.4.1. Supported Map Server Types

The Geoview feature can integrate with any map server that serves tiles using the XYZ tile standard, including:

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- **ArcGIS Server**: Enterprise GIS platform with tile service capabilities
- **OpenStreetMap tile servers**: Self-hosted OSM tile servers (e.g., using `mod_tile` and `renderd`)
- **GeoServer**: Open-source server for sharing geospatial data
- **MapProxy**: Open-source proxy for geospatial data
- **Any XYZ-compatible tile server**: Servers that provide tiles at URLs like `https://server.example.com/{z}/{x}/{y}.{format}`

11.4.2. Configuration Requirements

To connect to a self-hosted map server:

1. **Verify XYZ tile standard compatibility**: Ensure the map server serves tiles using the standard `{z}/{x}/{y}` URL pattern
2. **Network accessibility**: Confirm the map server is accessible from your Operations deployment
3. **Authentication**: Determine if the map server requires authentication (API keys, certificates, etc.)
4. **Tile formats**: Verify supported tile formats (PNG, JPG, WebP, etc.)

11.4.3. Integration Steps

1. **Contact the product team** to obtain configuration instructions for connecting to external map servers
2. **Provide map server details**:
   - Base tile URL template (e.g., `https://mapserver.customer.com/tiles/{z}/{x}/{y}.png`)
   - Authentication requirements
   - Supported zoom levels and geographic bounds
   - Any custom headers or parameters required
3. **Configure Operations** with the external map server settings through your deployment configuration

> **Note**
>
> External map server integration may require specific configuration parameters not covered in this import guide. Contact the product team for detailed integration instructions and compatibility verification.

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11.5. Step 3: Local Data Import Methods (Alternative to Self-Hosted Servers)

You can perform the import using either Kubernetes or Docker. Choose the method that best fits your environment.

> **Note**
>
> The import process requires an internet connection. The run process does not require an internet connection.

11.5.1. Method A: Kubernetes Import

11.5.1.1. Create Persistent Volume

Create the PersistentVolume to store the PostGIS database:

```yaml
apiVersion: v1
kind: PersistentVolume
metadata:
  name: osm-data
  labels:
    app: osm
    name: osm-data
spec:
  accessModes:
    - ReadWriteOnce
  capacity:
    storage: 10Gi
  hostPath:
    path: /mnt/osm-data
    type: DirectoryOrCreate
  persistentVolumeReclaimPolicy: Retain
  storageClassName: ""
```

11.5.1.2. Create Persistent Volume Claim

```yaml
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  name: osm-claim
spec:
  accessModes:
    - ReadWriteOnce
  storageClassName: ""
  resources:
```


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```
    requests:
      storage: 10Gi
  selector:
    matchLabels:
      app: osm
      name: osm-data
```

11.5.1.3. Create Import Pod

Create a temporary pod to perform the import operation:

```
apiVersion: v1
kind: Pod
metadata:
  name: import-osm
spec:
  containers:
  - name: osm-importer
    image: overv/openstreetmap-tile-server:2.3.0
    imagePullPolicy: IfNotPresent
    command: ["/bin/sh"]
    args: ["-c" , "/run.sh import"]
    env:
    - name: UPDATES
      value: "disabled"
    - name: ALLOW_CORS
      value: "enabled"
    volumeMounts:
    - mountPath: /var/lib/postgresql/12/main
      name: osm-data
    - mountPath: /data.osm.pbf
      name: osm-pbf-data
    resources:
      requests:
        memory: "2Gi"
        cpu: "1"
      limits:
        memory: "4Gi"
        cpu: "2"
    ports:
    - containerPort: 80
      name: http
      protocol: TCP
  restartPolicy: Never
  terminationGracePeriodSeconds: 30
  tolerations:
  - effect: NoExecute
    key: node.kubernetes.io/not-ready
    operator: Exists
    tolerationSeconds: 300
  - effect: NoExecute
    key: node.kubernetes.io/unreachable
```

```
    operator: Exists
    tolerationSeconds: 300
  volumes:
  - name: osm-data
    persistentVolumeClaim:
      claimName: osm-claim
  - name: osm-pbf-data
    hostPath:
      path: /host/osm-data/<region_name>.osm.pbf
      type: File
```

11.5.1.4. Deploy and Monitor

For k8 cluster environments, perform these steps before applying the Kubernetes manifests:

1. Copy your PBF file to the k8 cluster node:

```
scp /absolute/path/<region_name>.osm.pbf <node-user>@<node-ip>:/host/osm-data/<region_name>.osm.pbf
```

2. Verify the file is copied correctly:

```
ssh <node-user>@<node-ip> "ls -la /host/osm-data/"
```

3. Apply the Kubernetes manifests:

```
kubectl apply -f pv.yaml
kubectl apply -f pvc.yaml
kubectl apply -f import-pod.yaml
```

4. Monitor the import progress:

```
kubectl logs -f import-osm
```

5. After successful import, create and export the PostgreSQL data archive:

   Create a compressed archive of the imported PostgreSQL data from the osm-data persistent volume:

```
ssh <node-user>@<node-ip> "sudo tar -czf /tmp/osm-postgres-data.tar.gz -C /mnt/osm-data ."
```

   Verify the archive was created successfully:

```
ssh <node-user>@<node-ip> "ls -lh /tmp/osm-postgres-data.tar.gz"
```

   Copy the archive from k8 node to your host machine:

```
scp <node-user>@<node-ip>:/tmp/osm-postgres-data.tar.gz ./osm-postgres-data.tar.gz
```


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Verify the archive is available on the host machine:

```
ls -lh ./osm-postgres-data.tar.gz
```

11.5.2. Method B: Docker Import

For environments without Kubernetes, you can perform the import using Docker directly.

11.5.2.1. Create Docker Volume

First create a Docker volume to hold the PostgreSQL database that will contain the OpenStreetMap data:

```
docker volume create osm-data
```

11.5.2.2. Import from Local PBF File

Download an `.osm.pbf` extract from https://download.openstreetmap.fr/ for the region that you're interested in. You can then start importing it into PostgreSQL by running a container and mounting the file as `/data/region.osm.pbf`:

```
docker run \
    -v /absolute/path/to/<region_name>.osm.pbf:/data/region.osm.pbf \
    -v osm-data:/data/database/ \
    overv/openstreetmap-tile-server:2.3.0 \
    import
```

If the container exits without errors, then your data has been successfully imported and you are now ready to run the tile server.

11.5.2.3. Import with Automatic Updates (Optional)

If your import is an extract of the planet and has polygonal bounds associated with it, like those from https://download.openstreetmap.fr/, then it is possible to set your server up for automatic updates. Make sure to reference both the OSM file and the polygon file during the import process:

```
docker run \
    -e UPDATES=enabled \
    -v /absolute/path/to/<region_name>.osm.pbf:/data/region.osm.pbf \
    -v /absolute/path/to/<region_name>.poly:/data/region.poly \
    -v osm-data:/data/database/ \
    overv/openstreetmap-tile-server:2.3.0 \
    import
```


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> **Warning**
>
> If you're not importing the whole planet, then the `.poly` file is necessary to limit automatic updates to the relevant region. When you only have a `.osm.pbf` file but not a `.poly` file, you should not enable automatic updates.

11.5.2.4. Download and Import Automatically

It is also possible to let the container download files for you rather than mounting them in advance by using the `DOWNLOAD_PBF` and `DOWNLOAD_POLY` parameters:

```
docker run \
    -e DOWNLOAD_PBF=https://download.https://download.openstreetmap.fr//europe/<region_name>-
latest.osm.pbf \
    -e DOWNLOAD_POLY=https://download.https://download.openstreetmap.fr//europe/<region_name>.poly \
    -v osm-data:/data/database/ \
    overv/openstreetmap-tile-server:2.3.0 \
    import
```

11.5.3. Monitoring Import Progress

For Kubernetes deployments:

```
kubectl logs -f import-osm
kubectl describe pod import-osm
```

For Docker deployments:

```
docker logs <container-id>
```

11.5.3.1. Export Docker Volume for Air-Gapped Deployment

After successful import, export the Docker volume as a tar file for transfer to isolated systems:

1. Create a temporary container to access the volume data:

   ```
   docker run --rm -v osm-data:/data -v $(pwd):/backup alpine tar -czf /backup/osm-postgres-
   data.tar.gz -C /data .
   ```

2. Verify the archive was created:

   ```
   ls -lh osm-postgres-data.tar.gz
   ```

3. Transfer the `osm-postgres-data.tar.gz` file to your target air-gapped system.

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11.6. Step 4: Import Duration and Storage Requirements

- **Duration**: Import operations typically take minutes to hours depending on file size (up to 3 hours for 1 GB PBF files)
- **Storage**: The resulting PostGIS database is 10-100 times larger than the original PBF file
- **Resources**: Ensure adequate CPU and memory resources during import (minimum 2Gi RAM, 1 CPU core)

11.7. Step 5: Transfer to Air-Gapped Environment

After successful import:

1. Stop the import pod/container
2. Export the Kubernetes volume or copy the Docker volume data
3. Transfer the exported data to your air-gapped Kubernetes environment
4. Create equivalent PersistentVolume and PersistentVolumeClaim in the target environment
5. Deploy Operations with the OpenStreetMap tile server enabled

11.8. Step 6: Deploy Operations OSM Tile Server in Air-Gapped Environment

11.8.1. Restore Data in Air-Gapped Environment

1. Transfer the `osm-postgres-data.tar.gz` file to your air-gapped Kubernetes node:

   ```
   scp ./osm-postgres-data.tar.gz <node-user>@<node-ip>:/tmp/osm-postgres-data.tar.gz
   ```

2. SSH into the target air-gapped Kubernetes node and restore the PostgreSQL data:

   ```
   ssh <node-user>@<node-ip>
   ```

   Once connected to the k8s node, create the directory and extract the data:

   ```
   sudo mkdir -p /mnt/osm-data-restored
   sudo tar -xzf /tmp/osm-postgres-data.tar.gz -C /mnt/osm-data-restored
   sudo chown -R postgres:postgres /mnt/osm-data-restored
   ```

   Verify the data was extracted correctly:

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```
sudo ls -la /mnt/osm-data-restored
```

11.8.2. Deploy OpenstreetMap Tile Server from Operations Helm Chart

1. Update your Operations Helm chart value `osmTileServer.enabled: true` to deploy the OSM tile server with the restored data volume. Ensure your `values.yaml` includes:

```
featureFlags:
  geoview: true

osmTileServer:
  enabled: true
  volume:
    pv:
      path: <osm-data-restored-path> # example: /mnt/osm-data-restored
```

2. Deploy or upgrade the Operations Helm chart with the updated values, specifying the restored volume path:

```
helm upgrade --install operations ./operations-chart \
  --set osmTileServer.enabled=true \
  --set featureFlags.geoview=true \
  --set osmTileServer.volume.pv.path=/mnt/osm-data-restored \
  --namespace operations
```

11.8.3. Verify Deployment

1. Verify the tile server is running and has access to the OSM data:

```
kubectl get pods -l app=operations-osm-tile-server
kubectl logs -l app=operations-osm-tile-server
kubectl describe pvc operations-osm-tile-server-claim
```

2. The tile server should now be running with access to the imported OpenStreetMap data, enabling the Geoview feature in your air-gapped environment. To verify the deployment, you can port-forward the `operations-osm-tile-server` service to your local machine and make an HTTP request to fetch a map tile:

```
kubectl port-forward svc/operations-osm-tile-server 8080:80 -n operations
```

Then, from your host, run:

```
curl -v http://localhost:8080/tile/0/0/0.png --output tile.png
```


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If you receive a PNG file and a `200 OK` response, the tile server is working correctly.

Alternatively, if accessing from outside the cluster (after authentication), call:

```
curl -v https://<host>/tile/0/0/0.png --output tile.png
```

A successful response with a PNG tile confirms the deployment is operational.

11.9. Troubleshooting

11.9.1. Common Issues

- **Insufficient storage**: Ensure the persistent volume has adequate space (10-100x the PBF file size)
- **Memory limits**: Increase memory limits if the import fails due to OOM errors
- **Long import times**: Large PBF files (>1GB) may require several hours to import
- **File not found**: Verify the PBF file path and filename in the pod specification

11.10. Additional Resources

- OpenStreetMap Downloads
- OSM Convert Tool Documentation
- OpenStreetMap Tile Server Docker Image

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